

spaceThink — Project Dossier

spaceThink

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Phase 4 — Hypothesis layer: templates + LLM + gates (wk 6-9)

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Verification note

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 - 1.1 A Process-Centric Survey of AI for Scientific Discovery Through the EXHYTE Framework
2. Closed-loop AI-scientist systems: what works, what's criticized
 - 2.1 Coscientist — Autonomous chemical research with large language models
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3. LLM hypothesis-generation quality: benchmarks & evaluation
 - 3.1 HypoBench — systematic benchmarking of hypothesis generation
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 - 4.1 Exploring LLM-based Agents for Root Cause Analysis
 - 4.2 AIOpsLab — holistic evaluation framework for autonomous-cloud agents
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5. Human-AI collaboration & uncertainty communication for expert review

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What this means for our MVP

Verification note

Part IX — Research: Digital Twins & SBI

1. Open-source spacecraft simulators usable as a digital twin

1.1 Basilisk (AVS Lab, University of Colorado / LASP)

1.2 NASA 42 (Goddard)

1.3 NASA NOS3 — Operational Simulator for Space Systems (cFS + 42 + hardware models)

1.4 Orekit

1.5 poliastro (archived) → hapsira (fork)

2. Spacecraft digital twin literature and programs

2.1 Orion SysML Model, Digital Twin, and Lessons Learned for Artemis I (NASA)

2.2 Digital Twin of Space Environment: Development, Challenges, Applications, and Future Outlook

2.3 Digital twin construction from fault and behaviour trees (MODELS-C 2022)

2.4 Sedaro (commercial)

2.5 Antaris TrueTwin (commercial)

3. Simulation-based inference (SBI)

3.1 The sbi package (sbi-dev)

3.2 Fast Bayesian equipment condition monitoring via SBI (heat exchanger health)

3.3 BayesFlow — amortized Bayesian workflows and model comparison

3.4 SBI with NPE applied to X-ray spectral fitting (A&A)

3.5 SBI for structural system identification, model updating, and damage detection

3.6 Machine Learning for Reaction Wheel Fault Detection using Simulated Telemetry Data (CU Boulder, uses Basilisk)

4. Bayesian optimal experimental design and active learning

4.1 Pyro pyro.contrib.oed — EIG estimation framework

4.2 Deep Adaptive Design (DAD) — amortized sequential BOED

4.3 Optimal Bayesian design for model discrimination via classification

4.4 Bayesian input design for linear dynamical model discrimination

4.5 BoTorch — Bayesian optimization / active learning infrastructure

5. Fault injection frameworks and fault libraries

5.1 NASA VirtualADAPT — simulated EPS testbed with built-in fault injection

5.2 ADAPT dataset and Diagnostic Competition (DXC)

5.3 Reaction-wheel fault simulation literature (friction, stiction, encoder faults)

5.4 OPSSAT-AD — real satellite telemetry anomaly benchmark (ESA OPS-SAT)

5.5 cFS Test Framework (CTF) and flight-software fault injection

What this means for our MVP

Recommended simulator: Basilisk, with NOS3 as the later high-fidelity stage

Recommended SBI approach for hypothesis scoring: amortized NPE (sbi package) for fault parameters + neural model comparison for I

Simplest viable experiment-design method for v1: discrete-menu EIG via nested Monte Carlo (Pyro contrib.oed), with classifier-based m

Part X — Research: Market Landscape

1. Competitive landscape

1.1 The map

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2. YC precedent in space & AI-diagnostics

3. Market size & dynamics

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Done

Pending (next session, in order)

Failed / blocked

Key context for whoever picks this up

spaceThink — Project Dossier

An EXHYTE agent for autonomous scientific reasoning over deep-space telemetry.

Prepared: 2026-07-20 · **Status:** pre-MVP, strategic direction open

This dossier collects every planning and research artifact for the spaceThink project into one document. **Part I lays out the several directions the project can take** — from autonomous scientific discovery (the original vision) to a commercial anomaly-diagnosis wedge — and should be read first. Parts III–V detail the product specification, the phased implementation plan, and how to split that build across 2–4 people; they currently work one specific direction in depth while the others remain open framings of the same engine. Every research citation in Parts VII–X was verified against a visited source. The markdown sources remain the editable originals; this PDF is a build artifact regenerated from them on demand.

Part I — Strategic Directions

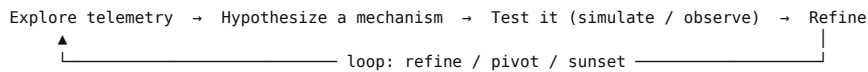
Strategic Directions

Status: open decision (2026-07-20). This document is deliberately **not** a commitment. It records the several directions spaceThink can take and what each one changes, so the founders can choose consciously rather than drift into one by default.

Read this first. Parts II–III of this dossier (the Specification and Implementation Plan) and the MVP Roadmap currently detail **Direction C (ops-first)** as the most fully-worked instantiation — because that is where the market research pointed for near-term customers and revenue. Directions A and B below are equally valid framings of the *same underlying system*; they are documented here but not yet specced to the same depth. Choosing A or B reshapes the users, data, demos, and success metrics in the later Parts, but not the core engine.

1. The invariant core (true in every direction)

Whatever we choose, the machine is the same — the EXHYTE closed loop:

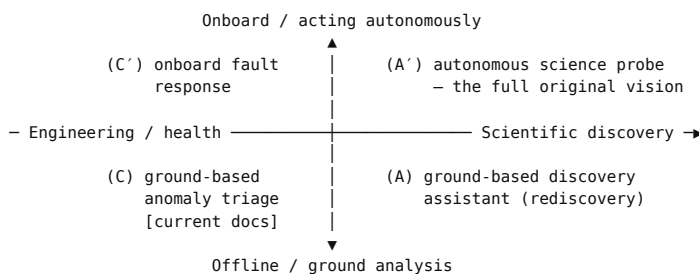


What never changes: the exploration/anomaly engine, the machine-readable hypothesis objects, the digital-twin testing layer, the simulation-as-truth gate (nothing unverified reaches a human), and the human-on-the-loop posture. **What changes between directions is the target of the hypotheses, and therefore who cares, what data feeds it, and how you prove it works.**

The bridge between “engineering” and “science” is one idea: an anomaly that is *neither a known failure mode nor a commanded maneuver* is exactly where new physics hides. The health machinery is what rules out the boring explanations so the genuinely novel ones surface. So science does not require a different engine — it is the ceiling built on the same floor.

2. Two axes, not one choice

The directions live on two independent axes:



- **Axis 1 — Purpose:** diagnose the spacecraft (engineering) ↔ discover things about the world (science).
- **Axis 2 — Autonomy:** analyze on the ground after the fact ↔ reason and act onboard under communication delay (re-point an instrument, schedule a follow-up).

Direction **B is the diagonal**: start bottom-left (C), move toward top-right (A') as data, customers, and trust accrue. The MVP stays in the bottom row (offline) for *all three* directions — onboard autonomy is a later-stage move gated by safety, and nothing about it needs to be decided now.

3. Direction A — Science-First (Autonomous Discovery)

Thesis. An AI scientist for spacecraft: the full discovery loop run over telemetry *and instrument data*, autonomously surfacing and testing hypotheses about real physical phenomena. AEGIS generalized — from images to telemetry-wide reasoning, and from target-selection to mechanism discovery. This is the closest match to the original vision: “robots and satellites doing closed-loop experiments that yield new scientific discoveries.”

- **Who cares / who pays:** planetary-science and heliophysics mission teams, instrument PIs, agency science directorates (NASA/ESA), science-operations centers. Revenue is grant- and mission-funded (NASA ROSES, science-topic SBIRs), not commercial SaaS.
- **Data:** science/instrument telemetry (magnetometers, plasma/particle detectors, spectrometers, radiation monitors) plus housekeeping for context; archival mission data (NASA PDS, CDAWeb/heliophysics, ESA PSA).
- **MVP demo — rediscovery:** take a dataset where a phenomenon was *later confirmed* (a known space-weather event, an instrument artifact that turned out to be real physics, a discovered periodicity), hold the answer out, and show the agent proposing and confirming it. This is evaluable and it directly shows the vision.
- **How you prove it works:** rediscovery rate on a curated set of known discoveries; novelty *and* truthfulness scoring (per TruthHypo — a novel-sounding claim that contradicts known physics is worse than useless); expert judgment.
- **Fundability:** hardest near-term revenue answer; strongest mission/ambition narrative. YC will love the ambition and press hard on “who pays this year.”
- **Hardest risk:** the oracle problem — what *counts* as a discovery, and how to evaluate it; long science feedback loops; agency sales cycles; instrument-data access.
- **Delta from current docs:** swap users/data/demos; keep the engine and the twin; add instrument-data ingestion and a novelty-vs-known-physics gate; success metrics become rediscovery-based, not fault-ranking.

4. Direction B — Science as Mission, Ops as Wedge (Recommended default)

Thesis. Same engine, sequenced. Commercialize health/anomaly diagnosis first (the wedge that has customers, data, and clean evaluation today) while the *stated mission and roadmap* is autonomous science. The ops product funds the moonshot and generates the operator relationships and telemetry access that the discovery capability later needs.

- **Who cares / who pays:** commercial operators now → science teams / agencies later.
- **Data:** housekeeping telemetry now (ESA-ADB, OPS-SAT-AD) → instrument/science data later.
- **MVP demo — both:** the ops triage / verified-root-cause demo **and** one rediscovery thread, so investors see the floor (revenue) *and* the ceiling (the mission). The discovery thread must be genuinely built, not a slide.
- **How you prove it works:** fault top-1/top-2 accuracy now, plus one rediscovery case.
- **Fundability:** strongest overall — a near-term revenue answer *and* a moonshot. This is how most credible deep-tech companies frame themselves (wedge product, mission ambition).
- **Hardest risk:** split focus and two customer types; discipline to keep the discovery thread real rather than letting the ops product quietly become the whole company (which is exactly the drift that produced the first version of this dossier).
- **Delta from current docs:** the current SPEC/Plan stand as *Phase 1 of B*; add the science north-star to the roadmap narrative and one instrument-data rediscovery demo to the MVP.

5. Direction C — Ops-First (Verified Anomaly Diagnosis)

Thesis. A verified anomaly-diagnosis product for fleet operations: detect → hypothesize the fault → verify it in the twin → recommend the next action. Scientific discovery stays in Future Work, exactly as the founding paper’s “Scope and assumptions” section scoped it. **This is what Parts II-III of this dossier currently specify.**

- **Who cares / who pays:** LEO constellation operators (5-50 sats, 3-15 person ops teams); GSaaS / ops-outsourcing providers as a channel; defense SBIR as a year-2 layer.
- **Data:** ESA-ADB, OPS-SAT-AD housekeeping telemetry.
- **MVP demo:** the two demos already in the roadmap (live ESA-anomaly closed loop; the muted-alarm triage demo).
- **How you prove it works:** true injected fault top-1 $\geq 60\%$ / top-2 $\geq 80\%$; detection within ± 0.05 event-wise F0.5 of the ESA-ADB SOTA baseline.
- **Fundability:** clearest customer and revenue wedge of the three.
- **Hardest risk:** it is not the vision the founders described; commoditization pressure from Intella, Quindar, Constellation Space, and adjacent well-funded players.
- **Delta from current docs:** none — it is the current plan.

6. Side-by-side

Axis	A — Science-first	B — Mission+Wedge	C — Ops-first
Primary user	Science/mission teams, agencies	Operators now → science later	Constellation operators
Data	Instrument + science telemetry	Housekeeping now → instrument later	Housekeeping telemetry
MVP demo	Autonomous rediscovery	Ops triage + one rediscovery	Ops triage / verified root cause
Evaluation	Rediscovery rate; novelty+truth	Fault accuracy + one rediscovery	Fault top-1/top-2; event-wise F0.5
Near-term revenue	Grants / mission-funded	Commercial SaaS (funds mission)	Commercial SaaS
Vision fidelity	Highest	High	Low
Hardest risk	Evaluating “a discovery”	Focus / real discovery thread	Not the vision; commoditization
Current spec depth	Framing only	Framing + C = phase 1	Fully specced (Parts II-III)

7. How to decide (not now, but soon)

- If a **revenue + YC answer within ~12 months is a hard constraint** → C, or B’s phase 1.
- If **the mission is the point** and you can fund it through grants/agency partnerships, and you’re willing to accept fuzzier evaluation → A.
- If you want **both** and are willing to sequence them → B (and treat the current Parts II-III as B’s first phase).
- **Autonomy (onboard, acting) is orthogonal** and is a later-stage move in every case; the MVP stays offline/replay regardless, for safety and tractability.

These are not permanently mutually exclusive. B is literally “C then A.” The one path to *avoid* is choosing C by accident — shipping the ops product and waking up in two years as a satellite-ops diagnostics vendor because the discovery mission never got a line in the plan. This document exists so that doesn’t happen silently.

8. Recommended next step

Pick a **primary** direction to detail next (rewrite the SPEC §2 goals, the roadmap, and the success metrics to match), while keeping this document as the standing record of the paths not taken. If undecided, default to **B**: it preserves the vision as the actual point of the company while keeping the fundable wedge already worked out in Parts II-III.

Part II — MVP Roadmap (YC Framing)

MVP Roadmap — YC Framing

Date: 2026-07-19. Companion to `docs/IMPLEMENTATION_PLAN.md` (build phases) and `docs/research/04-market-landscape.md` (full market evidence).

The pitch

Satellite fleets triple every five years, but anomaly response is still a war room. spaceThink is the AI flight-controller that detects an anomaly, hypothesizes the physical root cause, and **proves it against a digital twin** before an operator touches a console.

Fallback framing: “Verified root-cause-as-a-service for spacecraft — we don’t just flag anomalies, we close them.”

Why now (all 2024–2026 facts, cited in research/04):

- ~10.5–14.5K active satellites, 3× since 2019, 50–100K projected by 2030; manual ops break past ~5–10 sats.
- ViaSat-3 anomaly: \$420M insurance claim, largest in history; underwriters exiting.
- YC is actively funding LLM-diagnostics-on-machine-data (Telemetron, Deepttrace, Daqstra) and space ops (Quindar S22, Constellation Space W26). The intersection is unclaimed.
- The architecture just became publicly feasible (EXHYTE survey Dec 2025; AURA Nov 2025; amortized-SBI fault diagnosis 2026) — first credible vertical product wins the category.

The moat question (“why won’t Quindar/Intella do this?”): they stop at detection and workflow. Our loop is *generative* + *verified*: mechanistic hypotheses tested in simulation before anyone sees them. That requires physics/SBI + agent engineering they don’t have, and it produces the asset they can’t copy quickly — a **fault-signature library** (fault parameterizations + trained amortized scorers per subsystem) that compounds with every customer anomaly we close.

What we demo (two demos, both from the implementation plan’s Phase 5)

1. **Real ESA anomaly, closed live** — stream OPS-SAT benchmark telemetry; spaceThink detects the anomaly, generates 3 ranked mechanistic hypotheses, runs each in the Basilisk twin, and one visibly reproduces the real telemetry signature. Detection → verified root cause in <5 minutes, side-by-side with “human anomaly review board: days-to-weeks.”
2. **The muted-alarm demo** — a day of noisy telemetry, hundreds of threshold alerts; spaceThink triages ~98% as benign-with-explanation (telecommand-aware) and escalates the two real faults with twin-verified causes. This attacks the pain Intella’s own CCO says is the real cost sink: triage, not detection.

Honesty rule for both: metrics shown are event-wise F0.5 on ESA-ADB protocol and top-1/top-2 hypothesis accuracy on the fault-injection suite — no point-adjusted F1, ever. In a field notorious for inflated claims, evaluation rigor is itself a differentiator.

Timeline (12 weeks to demo-ready; two founders)

Weeks	Track A (data/physics)	Track B (inference/agent)	Milestone
1–3	Ingest + detector on OPS-SAT-AD	Runstore, Basilisk scenario + fault library	Detector reproduces published baselines
4–5	ESA-ADB Mission1 subset + metrics	Ensemble runner + synthetic fault suite	“First light”: ranked events on real ESA data
6–7	Detector hardening, telecommand FP kills	SBI scorers + calibration	True fault top-1 ≥60% on synthetic suite
8–9	Knowledge/RAG wrapper	Templates + LLM generator/critic/ranker + causal gate	Full loop closes end-to-end
10–12	Dashboard + demo scripts	Planner (EIG menu) + polish	Both YC demos run from clean checkout

Weekly integration on the runstore contract; borromeo gate green at every milestone.

Success metrics (what “MVP works” means — from SPEC QAs)

- Detection: within ± 0.05 event-wise F0.5 of Telemanom-ESA-Pruned on ESA-ADB M1 subset.
- Diagnosis: true injected fault top-1 $\geq 60\%$, top-2 $\geq 80\%$ (≥ 30 scenarios, ≥ 5 fault classes).
- Triage story: $>90\%$ of rare-nominal (telecommand-driven) events auto-explained on Mission2 annotations.
- Ops: full loop ≤ 5 min/event on a laptop; $\leq \$2$ LLM spend/run; dashboard replay instant.

First customers & GTM sequence

1. **Design partners (now):** 2–3 LEO constellation operators, 5–50 sats, 3–15 person ops teams (EO/IoT/comms, 2–5 yrs post first launch) — offer free pilot on their historical anomaly archive (“we’ll re-diagnose your last 10 anomalies”).
2. **Channel (post-batch):** GSaaS/ops-outsourcing providers (KSAT/Telespazio-type) — their margin is operator hours; one deal covers many fleets.
3. **Year 2:** defense non-dilutive layer (SpaceWERX/SDA SBIR — Sedaro’s proven path); insurance angle (root-cause evidence for claims/premiums) as a wedge with GEO operators.

Pricing sketch: per-satellite/month SaaS with a free triage tier (Kayhan freemium precedent), diagnosis/verification metered on the paid tier.

Investor objections → prepared answers

Objection	Answer
“Isn’t this Quindar/Intella/Constellation Space?”	They detect or automate workflow; nobody verifies hypotheses in simulation. The closed loop is the product.
“Where does the twin come from per customer?”	Fidelity-tiered: parametric Basilisk-class twins auto-calibrated from the customer’s own telemetry for common subsystems; partner/import path (Sedaro, TrueTwin) for high fidelity. The fault library transfers across customers even when twins don’t.
“TAM is small (~\$2B/yr ops software)”	Constellation growth (10K→50K+ sats), \$420M per-anomaly stakes + insurance-aligned pricing, and the same engine expands into industrial predictive maintenance (\$105B by 2035).
“LLM hallucination in safety-critical ops”	The twin is the hallucination filter — nothing unverified reaches an operator; causal-graph gate rejects impossible mechanisms pre-simulation; human-on-the-loop by default.
“Data access / ITAR”	Start with public ESA benchmarks + allied commercial operators under the relaxed 2024 EAR rules; cloud-hosted with data classification designed in from day one.

What we are NOT building for the batch (scope contract, SPEC §3)

No flight software, no real-time mission-scale streaming, no commanding real spacecraft, no multi-mission transfer learning, no instrument-science extension. Offline/replay + simulation only. Every one of these is a roadmap slide, not an MVP requirement.

Part III — Product Specification

SPEC — spaceThink: EXHYTE Agent for Autonomous Scientific Reasoning over Spacecraft Telemetry

Status: v0.2 — decisions finalized 2026-07-19 after literature review (docs/research/literature-review.md) Source: EXHYTE-research-doc.pdf (research context, gaps, proposed architecture) Method: CS130 — requirements separated from design; stories INVEST-validated; NFRs as quality-attribute scenarios; alternatives generated before decisions.

1. Context & Scope

Spacecraft generate high-dimensional multivariate telemetry under bandwidth/latency/compute constraints. Existing AI systems detect anomalies (scores/labels) but do not explain them: no mechanistic hypotheses, no testing against simulation, no closed discovery loop. The EXHYTE framework (Exploration → Hypothesis → Testing) names this gap.

Product: an agent that ingests spacecraft telemetry, surfaces events of interest, generates explicit machine-testable mechanistic hypotheses about root causes, tests them against a digital twin, and presents ranked, evidence-backed diagnoses to operators.

MVP context: demoable at a YC interview/batch — offline (post-facto) analysis of public benchmark telemetry, with a live dashboard walking through one full closed loop.

2. Goals

- G1: Full closed EXHYTE loop on ≥ 1 public dataset and ≥ 1 simulated subsystem.
- G2: Measurably better diagnosis than a detector-only baseline (time-to-diagnosis, hypothesis precision/recall on injected faults).
- G3: A demo a non-space investor understands in 3 minutes.

3. Non-Goals (MVP)

- On-board/flight deployment (radiation-hardened, FPGA) — future work.
- Real-time streaming ingestion at mission scale — offline/replay only.
- Multi-mission transfer learning; RL-based experiment design.
- Commanding real spacecraft. All “tests” run in simulation.
- General science-instrument data (spectrometers/imagers) — telemetry/health only.

4. Users & Personas

- **P1 Mission operator:** monitors fleet health, triages alerts, drowning in false alarms.
- **P2 Spacecraft systems engineer:** investigates anomalies post-facto; writes anomaly reports; owns the digital twin/physics knowledge.
- **P3 Founder/demo audience (YC):** needs to see the loop close live.

5. Functional Requirements (user stories + acceptance criteria)

US-1 INGEST TELEMETRY

As a systems engineer, I want to load a public telemetry dataset (channels, telecommands, labeled anomalies) into a common internal representation so that all downstream layers work against one schema.

- AC1: Given an ESA-ADB or SMAP/MSL archive on disk, when I run `spacethink ingest <path>`, then channels + events are stored in the internal store and `spacethink info` reports channel count, time range, and anomaly-label count.
- AC2: Given a malformed file, when ingested, then the CLI fails fast with a clear error naming the offending file/field (no partial silent import).

US-2 DETECT EVENTS OF INTEREST (EXPLORATION)

As an operator, I want the agent to rank anomalous/interesting telemetry segments so that I look at the right 1% of data first.

- AC1: Given an ingested dataset with labels held out, when detection runs, then the system emits ranked events with start/end, involved channels, and an anomaly score; event-wise recall $\geq$ baseline LSTM-AE reproduction on the same split.
- AC2: Given a detected event, then the event record includes context (active telecommands, mission phase) needed by hypothesis generation.

US-3 GENERATE MECHANISTIC HYPOTHESES (HYPOTHESIS)

As a systems engineer, I want explicit, parameterized candidate explanations for an event so that investigation starts from mechanisms, not raw scores.

- AC1: Given an event of interest, when hypothesis generation runs, then ≥ 2 distinct hypotheses are produced, each as a machine-readable record: template id, target subsystem, parameter(s) with prior ranges, natural-language rationale, confidence, assumptions.
- AC2: Every hypothesis maps to ≥ 1 executable simulator manipulation (schema-validated); hypotheses that cannot be simulated are flagged `untestable` with a reason.
- AC3: Given a hypothesis referencing a physically impossible cause (violates the system causal graph), then it is rejected before reaching the user.

US-4 TEST HYPOTHESES AGAINST DIGITAL TWIN (TESTING)

As a systems engineer, I want each hypothesis scored against simulation so that belief is grounded in physics, not LLM confidence.

- AC1: Given a testable hypothesis, when testing runs, then the simulator executes an ensemble over the hypothesis's parameter range and the system computes a fit score (likelihood/statistical distance) of real vs simulated telemetry.
- AC2: When all hypotheses for an event are tested, then output is a posterior-ranked list; on synthetic events with known injected faults, the true fault ranks #1 in $\geq 60\%$ of cases and top-2 in $\geq 80\%$ (MVP bar; tighten later).

US-5 PROPOSE NEXT-BEST OBSERVATION

As an operator, I want the agent to tell me what data/test would best discriminate the top hypotheses so that I know what to do next.

- AC1: Given ≥ 2 competing hypotheses with posterior overlap, then the system proposes a ranked list of discriminating observations/commands with expected information gain, constrained by a declared safety allowlist.

US-6 REVIEW IN DASHBOARD

As a founder, I want a dashboard that replays: telemetry $\rightarrow$ event flag $\rightarrow$ hypotheses $\rightarrow$ simulation traces vs real data $\rightarrow$ ranked diagnosis $\rightarrow$ suggested next observation.

- AC1: Given a completed analysis run, the dashboard renders the full loop for a selected event in < 3 clicks, with plots of real vs simulated channels per hypothesis.
- AC2: Every AI-generated claim shown displays its evidence (fit score, sim runs, retrieved docs) — no unexplained assertions.

US-7 REPRODUCIBLE RUNS

As a systems engineer, I want every analysis run captured (config, seeds, model+prompt versions, sim parameters) so that results are auditable and rerunnable.

- AC1: Given a run id, `spacethink rerun <id>` reproduces rankings within stochastic tolerance; run manifest is stored as a versioned artifact.

6. Non-Functional Requirements (quality-attribute scenarios)

- **QA-1 Correctness/graded trust:** On the synthetic fault-injection suite (≥ 30 scenarios, ≥ 5 fault classes), true-fault top-1 $\geq 60\%$, top-2 $\geq 80\%$; no `untestable` hypothesis ever presented as tested.
- **QA-2 Performance (demo):** Full loop on one event (detection $\rightarrow$ hypotheses $\rightarrow$ 10-sim ensemble $\rightarrow$ ranking) completes in ≤ 5 min on a laptop-class machine; dashboard replay is instant (pre-computed artifacts).
- **QA-3 Modifiability:** Swapping the anomaly detector, the LLM provider, or the simulator touches exactly one adapter module each (verified by a change-impact test list).

- **QA-4 Testability:** Every layer runnable headless with stub DOCs (stub LLM, stub simulator); CI green without network access or API keys.
- **QA-5 Safety:** Proposed observations/commands validated against a declarative allowlist; anything outside is marked `requires human approval` — enforced by schema, tested.
- **QA-6 Cost:** One full-loop analysis $\leq$ \$2 in LLM API spend at demo settings (cached, logged per run).

7. Architecture (drivers → decisions)

Drivers: closed-loop pipeline over heterogeneous data (functional); QA-3 swap-ability (detectors/LLMs/simulators are all likely to change — each is a module secret); QA-4 testability; demo-speed performance.

Style: Pipes & Filters for the analysis loop (Exploration → Hypothesis → Testing → Report), each stage a filter over typed artifacts persisted to a run store; Layered inside each stage (domain core, adapters at edges — hexagonal). Dashboard reads the run store only (never computes) — decouples demo from compute.

Module decomposition (information hiding — each hides one likely-to-change decision):

```
src/spacethink/
  domain/          # entities: Channel, EventOfInterest, Hypothesis, SimResult, RunManifest
                   # (pure, no I/O — the stable vocabulary every layer shares)
  ingest/          # SECRET: dataset formats (ESA-ADB vs SMAP/MSL parquet/csv quirks)
  explore/         # SECRET: detection algorithm (LSTM-AE now, anything later)
  knowledge/       # SECRET: retrieval strategy over mission docs/anomaly reports
  hypothesize/    # SECRET: LLM provider + prompt/template versions; causal-graph filter
  twin/           # SECRET: simulator choice (Basilisk) + fault-parameter mapping
  evaluate/       # SECRET: fit metric (likelihood vs distance) + posterior computation
  plan/           # SECRET: experiment-design method (info-gain v1)
  runstore/       # SECRET: artifact persistence layout (fs/sqlite now)
  cli/            # thin orchestration: compose filters into the pipeline
  dashboard/      # read-only viewer over runstore artifacts
  tests/          # mirrors src; stub LLM + toy simulator fixtures; fault-injection suite
```

Key contracts (stable interfaces; JSON-schema validated at boundaries):

- `Hypothesis`: {id, template_id, subsystem, parameters: {name, prior_range, unit}, rationale, assumptions[], testable: bool, sim_mapping: {model, param_overrides}} — units are explicit everywhere (Mars Climate Orbiter rule).
- `SimResult`: {hypothesis_id, ensemble: [...], fit_score, method, uncertainty}.
- Every artifact carries `run_id` + producer version.

Alternatives considered

- *Agent-framework-first (LangGraph/CrewAI orchestrating everything)*: rejected for MVP core — hides control flow, hurts reproducibility (US-7) and testability (QA-4); LLM use is confined to `hypothesize/` + `knowledge/` behind interfaces.
- *Monolithic notebook prototype*: rejected — fine for experiments (keep in `notebooks/`), but the demo needs the deterministic pipeline + dashboard.
- *Streaming-first (Kafka etc.)*: rejected — Non-Goal; offline replay satisfies all MVP ACs.
- *Build own simulator*: rejected — Basilisk chosen (see §8); NASA 42 rejected (C-only, socket IPC, non-OSI NOSA license), NOS3 deferred (too heavyweight for ensembles), Orekit/hapsira rejected (no subsystem models); we own only the fault-parameter mapping.

8. MVP scope decisions (finalized — rationale in docs/research/literature-review.md §3)

1. **Datasets:** OPS-SAT-AD for iteration → ESA-ADB Mission1 lightweight subset (channels 41–46) as headline benchmark; SMAP/MSL smoke-test only. Reuse ESA-ADB's MIT evaluation pipeline verbatim; event-wise F0.5 / alarming precision / ADTQC; PA-F1 is banned.
2. **Simulator:** Basilisk (ISC, Python-native, Monte Carlo dispersions, reaction-wheel friction/jitter + EPS models). NOS3 reserved for later high-fidelity validation. Thermal faults out of v1 scope.
3. **Subsystem slice:** ADCS/reaction wheels first (fault priors: 5–20× static friction, randomized onset); EPS second (ADAPT/VirtualADAPT taxonomy: incipient/abrupt/bias/stuck).

4. **Baseline detector:** channel-wise LSTM/GRU forecaster + nonparametric dynamic thresholding (Telemanom lineage — current ESA-ADB SOTA), telecommand-conditioned from day one. No foundation models, no GAF encodings in v1.
5. **Hypothesis scoring:** amortized neural posterior estimation (`sbi`) for fault parameters within a family + amortized Bayesian model comparison across hypotheses; simulation-based calibration + posterior-predictive checks as guardrails.
6. **Experiment design v1:** discrete-menu expected-information-gain (Pyro `contrib.oed`) or classifier-based model discrimination; Deep Adaptive Design deferred to v2.
7. **Hypothesis layer:** EXHYTE substages (E1-E3, H1-H2, T1-T2) as an explicit state machine with refine/pivot/sunset transitions; role-separated Generator → Critic → tournament Ranker; LLM sees pre-digested features, never raw numeric series; causal-graph truthfulness gate before anything reaches simulation.

9. Risks

- R1 (highest): digital-twin/telemetry misalignment — sim never fits real data well enough to discriminate hypotheses. Mitigation: start with synthetic events injected via the same simulator (guaranteed alignment), then bridge to real data; report calibration honestly.
- R2: LLM hallucinated mechanisms → mitigated by template constraint + causal-graph filter + simulation gate (nothing untested reaches the ranked list).
- R3: benchmark label quality — resolved: ESA-ADB/OPS-SAT-AD primary, event-wise F0.5, PA-F1 banned (literature-review.md §2).
- R4: scope creep toward flight software — Non-Goals section is the contract.
- R5: competitive window — the architecture is published (AURA) and adjacent players are well-funded (Sift, Constellation Space); ship the vertical slice fast.
- R6: twin-provenance objection (“where does the twin come from per customer?”) — develop the fidelity-tiered answer: parametric twins auto-calibrated from telemetry + partnerships.

Part IV — Implementation Plan

Implementation Plan — spaceThink MVP

Date: 2026-07-19. Derives from `SPEC.md` (requirements + finalized decisions) and `docs/research/literature-review.md`.
Companion: `docs/MVP_ROADMAP.md` (YC framing/timeline).

Guiding constraints (CS130): each phase ships a vertically testable increment; every module hides one likely-to-change decision; detectors own numbers, LLMs own mechanism; nothing untested in simulation reaches a user; the borromeo gate must be green at every phase end.

Phase 0 — Foundations (DONE / in progress)

- ☒ Repo scaffold, `src/spacethink/domain` immutable entities, pytest, borromeo gate green.
- ☐ `runstore/`: content-addressed artifact store (filesystem + SQLite index). Every stage reads/writes typed artifacts keyed by `run_id`. This is US-7 (reproducibility) and what makes the dashboard trivially demoable (pre-computed replay).
- ☐ CI: gate == CI. Stub-only tests (QA-4): no network, no API keys, no Basilisk needed for unit tier (a `ToySimulator` implements the same `Twin` interface).

Exit criteria: `spacethink run --help` works; a dummy pipeline writes/reads artifacts; gate green.

Phase 1 — Exploration layer: ingest + detect + evaluate (wk 1-3)

Vertical slice on **OPS-SAT-AD** (MBs, 30 published baselines), then **ESA-ADB Mission1 lightweight subset (channels 41-46)**.

1. `ingest/`: parsers → `Channel` /segment artifacts. Schema-validate at boundary; fail fast with file/field-level errors (US-1 AC2). Telecommands are first-class inputs.
2. `explore/`: channel-wise GRU/LSTM one-step forecaster + **nonparametric dynamic thresholding**, error smoothing, detection merging, pruning (Telemanom lineage — thresholding/postprocessing gets equal engineering budget to the net). Telecommand conditioning from day one (exogenous inputs).
3. `evaluate/` (detection half): wrap the MIT-licensed ESA-ADB metrics implementation — corrected event-wise F0.5, alarming precision, ADTQC. Internal scalar: PR-based/PA%K. **PA-F1 is banned in code**: the metrics module simply does not implement it.

Exit criteria: reproduce Telemanom-class results on OPS-SAT-AD vs published baselines; event ranking with context lands in runstore (US-2 both ACs); gate green.

De-risk note: this phase is deliberately “boring reproduction” — it buys credibility (same benchmark, same metrics the field trusts) and produces the events the rest of the loop consumes.

Phase 2 — Digital twin + fault library (wk 2-5, overlaps Phase 1)

1. `twin/`: `Twin` interface (`configure(sim_mapping) → run(duration, seed) → channels`). Implementations: `ToySimulator` (test double, milliseconds) and `BasiliskTwin` (ISC license; scripted scenario: 3-axis spacecraft, 4 reaction wheels, star tracker, gyro, simple EPS).
2. Fault library v1 (ADCS): reaction-wheel friction multiplier (literature prior 5–20× static friction, randomized onset), wheel-speed encoder fault, gyro bias drift, sensor noise inflation. Each fault = a `SimMapping` template with bounded parameter ranges — this file IS the company’s proprietary asset seed.
3. Ensemble runner: Monte Carlo dispersions over fault parameters (Basilisk native), seeds recorded in `RunManifest`; parallelized via process pool; outputs standardized telemetry windows.
4. **Synthetic eval suite** (QA-1 backbone): ≥30 scenarios × ≥5 fault classes of Basilisk-generated telemetry with known injected faults. This is the ground truth for every downstream claim (hypothesis top-1/top-2 accuracy).

Exit criteria: `spacethink twin run --fault rw_friction --magnitude 8x` produces plausible faulty telemetry; ensemble of 100 runs < 10 min on laptop; suite artifacts in runstore; gate green.

Phase 3 — Testing layer: SBI scoring (wk 4-7)

1. `evaluate/` (inference half): amortized **Neural Posterior Estimation** via `sbi` (Apache-2.0) with an embedding network over telemetry windows — train offline per fault family on Phase-2 ensembles; inference in ms at run time (heat-exchanger precedent: 82x vs MCMC).
2. Cross-hypothesis scoring: amortized Bayesian model comparison (neural classifier over competing twin configurations → posterior model probabilities); `sbi`'s NRE as fallback.
3. Guardrails (non-negotiable): simulation-based calibration + posterior-predictive checks in CI for each trained scorer; a hypothesis whose posterior fails calibration is marked `low-confidence` in the artifact, and the UI must show it.
4. Sim-to-real bridge (top risk R1): validate scorers on OPS-SAT-AD labeled anomalies and ADAPT EPS fault data; report honest calibration numbers — this is a *feature* for credibility, not a weakness to hide.

Exit criteria: US-4 ACs — on the synthetic suite, true fault ranks top-1 $\geq 60\%$ / top-2 $\geq 80\%$; calibration diagnostics stored per run; gate green.

Phase 4 — Hypothesis layer: templates + LLM + gates (wk 6-9)

1. `hypothesize/templates.py`: parametric templates per fault family (“RW-N friction elevated by $k\times$ ”, “gyro bias drift at r°/h ”, ...) with priors from the fault library.
2. Causal graph: small typed DAG of the demo spacecraft (subsystem topology + which faults can produce which channel signatures). The **truthfulness gate**: LLM output that violates the graph is rejected pre-simulation (US-3 AC3).
3. LLM roles (role-separated, co-scientist pattern), all behind one `LlmClient` interface (QA-3; stub implementation for CI):
 - **Generator**: event features (pre-digested — never raw numeric series) + retrieved context → instantiated templates + free-form candidates with confidence + assumptions.
 - **Critic**: physical-plausibility and evidence review; per-mechanistic-step uncertainty.
 - **Ranker**: pairwise tournament over surviving candidates → prioritized queue for simulation budget (EXHYTE H2 — the field’s weakest substage, our differentiator).
4. `knowledge/`: RAG over mission docs/anomaly reports. **Buy, don’t build** — thin wrapper over an off-the-shelf embedding + retrieval stack; only the interface is ours.
5. Hypothesis quality eval (HypoBench framing): on the synthetic suite, measure hypothesis discovery rate — did the true generative mechanism appear in the candidate set?

Exit criteria: US-3 all ACs; end-to-end event → ranked verified hypotheses on synthetic suite meeting QA-1; LLM spend $\leq$ \$2/run logged (QA-6); gate green.

Phase 5 — Planner + dashboard + demo assembly (wk 8-12)

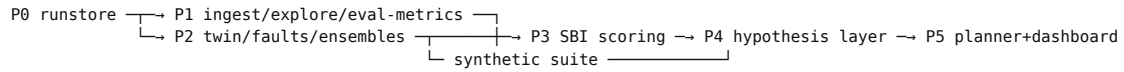
1. `plan/`: discrete design menu {keep observing, high-rate downlink channel set A/B, diagnostic slew, wheel spin-up/down test} scored by EIG (Pyro `contrib.oe` nested-MC; fallback: Hainy classifier discrimination — reuses Phase-2 ensembles as-is). Safety allowlist schema enforced (QA-5): out-of-list actions marked `requires human approval`.
2. `dashboard/`: read-only web app over runstore artifacts (FastAPI + static frontend). Replay flow: telemetry → event flag → hypotheses w/ per-step uncertainty → sim-vs-real overlay per hypothesis → posterior ranking → next-best observation. Every claim shows its evidence (US-6 AC2). Pre-computed artifacts → instant demo (QA-2).
3. Demo hardening: the two YC demos (see MVP_ROADMAP.md) scripted as reproducible runs; `spacethink rerun <id>` (US-7).

Exit criteria: full-loop demo < 5 min live / instant replay; both demo scripts run from a clean checkout; gate green.

Cross-cutting rules

- **Testing**: TDD per feature; stub DOCS (`StubLlm`, `ToySimulator`) keep CI hermetic; the synthetic fault suite is the regression bed for scientific claims, not just code.
- **Module boundaries** (QA-3 change-impact list): swapping detector / LLM provider / simulator touches exactly `explore/` / `hypothesize/llm_client` / `twin/` respectively — revalidate this list at each phase end.
- **Security**: no keys in code; LLM spend caps; safety allowlist schema-enforced.
- **Two-founder split** (Conway-friendly): one owns Exploration+Twin (data/physics track, Phases 1-2), the other owns Testing+Hypothesis (inference/agent track, Phases 3-4); integrate weekly on the runstore contract.

Dependency graph



P1 and P2 are parallel tracks (one per founder). P3 needs P2's ensembles; P4 needs P1's events + P3's scorers; P5 needs everything but is mostly presentation.

Part V — Parallel Work Split (2-4 People)

Parallel Work Split

How to divide the build across 2, 3, or 4 people so they work concurrently instead of serially. Companion to `docs/IMPLEMENTATION_PLAN.md` (the phases) — this document assigns the phases to people.

Grounding principle (CS130 / Parnas / Conway): **a module is an independent work assignment**. The architecture already hides each likely-to-change decision behind one module, so the team boundaries should follow those seams. Align people to interfaces, freeze the interfaces first, and everyone can build behind a stub without waiting on anyone else.

Direction note: this split is written for the currently-specced path (Direction C / phase 1 of B in `docs/STRATEGIC_DIRECTIONS.md`). §7 covers how it shifts for Direction A (science).

1. The one rule that makes parallel work possible

Freeze the shared contracts in Week 0, before splitting. Three things are touched by everyone; if they drift, every track breaks:

1. `domain/` entities (already built) — the shared vocabulary.
2. `runstore/` artifact schemas — how stages hand work to each other.
3. The three swap-interfaces — `Detector`, `Twin`, `LlmClient`.

Once these exist with **stubs on both sides**, each person builds their real implementation against the other side's stub. Nobody blocks. The stubs are not throwaway — they are the permanent test doubles that keep CI hermetic (no network, no Basilisk, no API keys).

2. The interface contracts (build these first, together)

Small, explicit, and frozen early. Signatures (Python-ish) — the real work hides behind them:

```
# domain/ — already implemented (immutable entities)
EventOfInterest, Hypothesis, FaultParameter, SimMapping, RunManifest

# runstore/ — the integration seam. Everyone reads/writes typed artifacts by run_id.
class RunStore:
    def put(self, run_id: str, kind: str, artifact) -> ArtifactRef
    def get(self, ref: ArtifactRef) -> artifact
    def list(self, run_id: str, kind: str) -> list[ArtifactRef]

# explore/ — Track A owns; Track B consumes events
class Detector(Protocol):
    def detect(self, channels, telecommands) -> list[EventOfInterest]
# stub: ThresholdDetector (trivial, deterministic) for other tracks' tests

# twin/ — Track A owns; Track B's SBI trains on its ensembles
class Twin(Protocol):
    def configure(self, mapping: SimMapping) -> "Twin"
    def run(self, duration_s: float, seed: int) -> Channels
# stub: ToySimulator (analytic ODE, milliseconds) — lets Track B start before Basilisk

# hypothesize/ — Track B owns
class LlmClient(Protocol):
    def generate(self, event_features, context) -> list[Hypothesis]
    def critique(self, hyp: Hypothesis) -> Critique
# stub: StubLlm (returns fixed templated hypotheses) — keeps CI offline

# evaluate/ — Track B owns
class Scorer(Protocol):
    def score(self, hyp: Hypothesis, real: Channels) -> SimResult # posterior/fit
```

Definition of “an interface is done”: a contract test exists that any implementation (real or stub) must pass. The contract test — not the implementation — is the source of truth both tracks code against.

3. Two people (the founders) — primary plan

Split along the deepest seam in the system: **numbers/physics** vs **reasoning/agent**. Each founder owns a vertical half of the loop end to end.

	Track A — Data & Physics	Track B — Inference & Agent
Owns modules	<code>ingest/</code> , <code>explore/</code> , <code>twin/</code> , fault library, synthetic suite, <code>evaluate/</code> (detection metrics)	<code>evaluate/</code> (SBI scoring), <code>hypothesize/</code> , <code>knowledge/</code> , <code>plan/</code>
Phase 0 (shared)	Co-build <code>domain</code> + <code>runstore</code> + interfaces + CI (pair or split 1 day)	same
Phase 1	Ingest OPS-SAT-AD/ESA-ADB; forecaster + dynamic thresholding; event-wise metrics	Scaffold <code>hypothesize/</code> + <code>evaluate/</code> against <code>ToySimulator</code> + <code>StubLLM</code> stubs
Phase 2	Basilisk scenario + fault library + ensemble runner + synthetic fault suite	Build SBI training/inference plumbing on toy ensembles; calibration harness
Phase 3	Hand real ensembles to B; help validate sim-vs-real on OPS-SAT/ADAPT	Train amortized NPE + model comparison on A's ensembles; SBC/PPC guardrails
Phase 4	Provide fault-parameter priors + causal-graph facts	LLM roles (generate/critic/rank), templates, causal-graph truthfulness gate, RAG
Phase 5	Detector polish; demo data prep	<code>plan/</code> (EIG menu); dashboard + demo scripts (or split with A)
Weekly sync	runstore contract test must stay green — the one meeting that matters	same

The single real cross-track dependency: Track B's SBI scorers (Phase 3) need Track A's Basilisk ensembles (Phase 2). Mitigation is built in — B works against `ToySimulator` from day one and only swaps in Basilisk ensembles when A ships them, so B is never idle. A should ship a crude "v0 ensemble" early (even 100 low-fidelity runs) to unblock B's real training sooner.

Shared/either-owner work: `cli/` (orchestration) and `dashboard/` (read-only viewer) sit on top of the runstore and can be built by whoever has slack, or split (A takes CLI, B takes dashboard — dashboard is closer to B's hypothesis/uncertainty presentation).

4. Three people

Split Track A's physics from its data, because they're different skill sets:

- **Person 1 — Data/Detection:** `ingest/`, `explore/` (forecaster + thresholding), detection metrics. Deliverable: ranked events in the runstore.
- **Person 2 — Twin/Physics:** `twin/` (Basilisk), fault library, ensemble runner, synthetic suite, **and** the SBI scoring in `evaluate/` (natural home — the person who owns the simulator owns the inference that consumes it). Deliverable: verified fault posteriors.
- **Person 3 — Agent/Product:** `hypothesize/` (LLM roles, templates, causal gate), `knowledge/`, `plan/`, `dashboard/`, `cli/`. Deliverable: the closed loop + the demo.

Interfaces between them are exactly the contracts in §2, so the coordination cost stays low. Critical path is Person 2 (twin → suite → SBI feeds both the detector's context and the agent's verification). Person 2 is the one to unblock first; if you hire one contractor, hire to relieve Person 2.

5. Four people

Add a dedicated inference owner and a dedicated product owner:

- **Data/Detection** (as above).
- **Twin/Simulation:** `twin/`, fault library, ensembles, synthetic suite.
- **Inference/SBI:** `evaluate/` — NPE, model comparison, calibration. Consumes the twin's ensembles; the cleanest stand-alone research role (portable ML skills, no Basilisk needed past the ensemble contract).

- **Agent/Product:** `hypothesize/`, `knowledge/`, `plan/`, `dashboard/`, `cli/`.

Beyond 4, parallelism stops paying: the loop only has so many independent seams, and coordination cost (Brooks) starts to dominate. Scale by deepening a track (more fault classes, more detectors, more subsystems), not by adding people to the same module.

6. Parallel schedule (2-person, 12 weeks)

Wk	Track A (Data & Physics)	Track B (Inference & Agent)	Integration
0	— co-build domain + runstore + interfaces + CI (both) —		contracts frozen
1	ingest OPS-SAT-AD	<code>hypothesize/</code> + <code>evaluate/</code> on stubs	stub loop runs e2e
2	forecaster + thresholding	SBI plumbing on ToySimulator	—
3	Basilisk scenario + first faults	calibration harness	v0 ensemble handoff
4	ESA-ADB subset + metrics	NPE trains on v0 ensemble	first real scores
5	fault library + synthetic suite +	model comparison across hyps	suite = shared truth
6	detector hardening (telecommand FPs)	SBC/PPC guardrails	top-1 ≥60% checkpoint
7	sim-vs-real on OPS-SAT/ADAPT	scorer validation on real data	calibration report
8	fault priors + causal-graph facts	LLM generate/critic/rank + gate	full loop closes +
9	demo data prep	templates + RAG (<code>knowledge/</code>)	—
10	→ dashboard (or CLI)	<code>plan/</code> EIG menu	—
11	demo script 1 (ops triage)	demo script 2 (+ rediscovery if B)	both demos green +
12	— harden, rehearse, buffer (both) —		gate green, demo-ready

♦ = milestone with a hard exit criterion (see IMPLEMENTATION_PLAN.md phase gates).

7. If the direction changes (A / science-first)

Direction A adds one workstream and reshapes evaluation, but keeps the same seams:

- **New stream — Science Data & Rediscovery** (a person or a shared responsibility): ingest instrument/science telemetry (magnetometer/plasma/spectrometer archives), and build the **rediscovery evaluation harness** (curated known-discovery datasets with answers held out). This replaces the fault-injection suite as the source of ground truth.
- Track A's `twin/` gains environmental/physical models (not just fault models); Track B's `hypothesize/` gains the novelty-vs-known-physics gate.
- The 2-person split still holds: A owns data+physics(+science data), B owns inference+agent(+rediscovery scoring). It's the same two halves, aimed at discovery instead of diagnosis.

8. Anti-patterns to avoid

- **Two people editing `domain/` or the runstore schema at once** → the seam drifts and both tracks break. Changes to shared contracts are a scheduled, agreed event, not a casual edit.
- **Blocking on a real implementation when a stub exists** → if you're waiting on Basilisk or the LLM, you skipped your stub. The stub is the point.
- **Letting the synthetic suite live in one person's head** → it's the shared definition of "does the science work"; it belongs in `tests/` in the runstore, owned jointly.
- **Adding people to a blocked track** → Brooks's Law. Unblock the critical path (the twin) first; only then does a new hand help.

Part VI — Literature Review: Synthesis

Literature Review — Synthesis

Date: 2026-07-19. Synthesizes the co-founder's research document ([EXHYTE-research-doc.pdf](#)) with four verified deep-dives (all citations checked against visited URLs):

- [01 — Telemetry anomaly detection SOTA](#) (22 sources)
- [02 — LLM agents for scientific discovery](#) (19 sources)
- [03 — Digital twins & simulation-based inference](#) (25+ sources)
- [04 — Market & YC landscape](#) (24 sources)

1. The original paper's thesis survives review — and sharpens

The co-founder's document claimed a gap: telemetry AI detects but doesn't explain; science autonomy (AEGIS) picks targets but doesn't reason system-wide; nobody closes the Exploration → Hypothesis → Testing loop. The review **confirms the gap is real in both the literature and the market**, and adds precision the original document didn't have:

1. **EXHYTE is a real, citable framework** (Hasib et al., Dec 2025, PMC12776506) with substages E1-E3 / H1-H2 / T1-T2 and three feedback routes (refine / pivot / sunset). Its own maturity assessment says EXPLORE is mature, **hypothesis prioritization (H2) is underdeveloped, and TEST lags the most** — our product lives exactly in the two weakest (least commoditized) stages.
2. **The exact architecture has been validated in an adjacent domain:** AURA (arXiv, Nov 2025. does LLM + digital twin + human-in-the-loop anomaly diagnosis for underwater robotics, and a 2026 heat-exchanger paper does amortized SBI fault diagnosis on a physics simulator with 82× speedup over MCMC. Feasibility is de-risked; **speed to the spacecraft vertical is the race**.
3. **Nobody in the market closes the loop:** funded players detect (Intella Mercury, Constellation Space YC W26), automate workflow (Quindar YC S22, Leanspace), or simulate (Sedaro, Antaris TrueTwin). Simulator-verified root cause is unclaimed territory.

2. Corrections to the original document

The review contradicts or refines several assumptions in the source paper:

- **SMAP/MSL are no longer credible headline benchmarks.** The paper lists them as candidate datasets; the field now considers them broken (trivial anomalies, mislabeled ground truth, point-adjustment F1 inflation — random scores can look SOTA). Use only as a smoke test. **ESA-ADB (2024) + OPS-SAT-AD (2025) are the credible benchmarks.**
- **GAF image encodings** (cited favorably in the paper) underperform badly on ESA-ADB (48.6% vs 92.7% CEF0.5 for forecasting+threshold). Drop from the plan.
- **"LLM-based reasoning over telemetry" needs a hard constraint the paper only implies:** LLMs fed raw numeric time series lose to statistical baselines (arXiv 2601.12448). The winning pattern is hybrid — detectors own numbers, LLMs own mechanism over pre-digested features (DML-LLM: F1 0.59→0.83, diagnosis time 7.4h→1.2h).
- **F1-style scores the paper references should be replaced** by ESA-ADB's operator-centric protocol: corrected event-wise **F0.5** (precision-weighted — false alarms are the #1 adoption killer), alarming precision, ADTQC timing. PA-F1 is banned in our reporting.

3. The six pending SPEC decisions — now decided

#	Decision	Choice	Basis
1	Dataset	OPS-SAT-AD first (tiny, CC BY 4.0, 30 baselines) → ESA-ADB Mission1 lightweight subset (ch. 41-46) as headline; SMAP/MSL smoke-test only	Report 01 \$7-8
2	Simulator	Basilisk (ISC license, Python-native, 365× realtime, Monte Carlo dispersions, RW friction/jitter + EPS models; Carneiro 2022 precedent). NOS3 later for high-fidelity validation; thermal out of v1 scope	Report 03 \$1, \$5.3
3	Subsystem slice	ADCS / reaction wheels (literature-standard fault parameterization: 5-20× static friction, randomized onset); EPS second (ADAPT dataset + VirtualADAPT fault taxonomy: incipient/abrupt/bias/stuck-at)	Report 03 \$5
4	Baseline detector	Channel-wise LSTM/GRU forecaster + nonparametric dynamic thresholding (Telemanom lineage — still SOTA on ESA-ADB, telecommand-conditioned from day one ; reuse ESA-ADB's MIT eval pipeline verbatim)	Report 01 \$3, \$8
5	Hypothesis scoring	Amortized NPE (sbi , Apache-2.0) for fault-parameter posteriors within a fault family + amortized Bayesian model comparison (BayesFlow-style classifier / NRE) across hypotheses; simulation-based calibration + posterior-predictive checks as guardrails	Report 03 \$3
6	Experiment design v1	Discrete-menu EIG (Pyro <code>contrib.oed</code> nested-MC) or the even simpler Hainy et al. classifier-discrimination method; Deep Adaptive Design is the v2 upgrade	Report 03 \$4

4. Agent-architecture findings (Hypothesis layer design)

From the AI-scientist literature (report 02), the patterns that work and the failures to design against:

- **Role separation wins:** Generator / Reflection-critic / Ranker / Evolver (Google co-scientist, FutureHouse Robin). Make prioritization (H2) an explicit scored tournament — the field's best answer to its weakest substage.
- **Sakana's AI Scientist is the failure catalog:** shallow retrieval → false novelty; 42% of experiments failed on code errors; hallucinated numbers in output. Countermeasures: truthfulness gate against a domain knowledge graph (subsystem topology + known failure modes, per TruthHypo), and **every claim bound to an executable check**.
- **Hypotheses must be structured executable objects, not prose:** {mechanism, affected signals, predicted checkable consequence, concrete test/sim to run, pass/fail predicate, evidence links} — this is already our `Hypothesis` domain entity; the review validates it.
- **Uncertainty must be per-mechanistic-step**, not one confidence number (coarse confidence causes automation bias; fine-grained uncertainty directs expert verification).
- **Buy EXPLORE, build TEST:** literature RAG is commoditized (PaperQA2); our engineering budget belongs in the Testing layer, which is both the field's weakest stage and the moat.
- **The simulation gate doubles as the hallucination answer** for investors: no hypothesis reaches an operator unless it reproduces the telemetry signature in the twin.

5. Market synthesis (YC positioning)

- **Wedge = triage + verified root cause**, not detection. Intella's own CCO concedes operators drown in benign alerts and mute alarms past 5-10 satellites.
- **First customer:** LEO constellation operator, 5-50 sats, 3-15 person ops team, 2-5 years post first launch. Channel: GSaaS providers (KSAT/Telespazio-type — one deal, many fleets). Defense SBIR = year-2 non-dilutive layer.
- **Demand drivers:** ~10.5-14.5K active satellites (3× since 2019, 50-100K projected by 2030); ops staffing curve breaks past ~5-10 sats; ViaSat-3 anomaly = \$420M claim (largest in history) with insurers exiting the market.
- **YC precedent is current:** Epsilon3 (S21), Quindar (S22), Constellation Space (W26, \$6.5M seed), plus F25/26 LLM-diagnostics bets (Telemetron, Daqstra, Deepttrace). The thesis is believed; the differentiation question ("why won't Quindar/Intella do this?") must be answered with the closed verified loop.
- **Pitch:** "Satellite fleets triple every five years but anomaly response is still a war room — the AI flight-controller that detects an anomaly, hypothesizes the physical root cause, and *proves it against a digital twin* before an operator touches a console."

6. Revised risk picture

Risk (SPEC §9)	Post-review status
R1 sim-to-real misalignment	Still the top technical risk, but mitigations are now concrete: heat-exchanger SBI precedent, Carneiro 2022 (Basilisk-trained fault detectors), OPS-SAT-AD + ADAPT as real-data validation sets, SBC/posterior-predictive calibration checks
R2 LLM hallucinated mechanisms	Mitigation upgraded: template constraint + causal-graph truthfulness gate + simulation gate + per-step uncertainty
R3 benchmark label quality	Resolved: ESA-ADB/OPS-SAT-AD primary; PA-F1 banned
R4 scope creep	Unchanged (Non-Goals hold)
R5 (new) competitive window	AURA proves the architecture publicly; Sift (\$67M) and Constellation Space are adjacent and well-funded. Window is open but closing — ship the vertical slice fast
R6 (new) twin-provenance objection	Investors will ask “where does the twin come from per customer?” Answer to develop: fidelity-tiered twins — Basilisk-class parametric twins auto-calibrated from telemetry + partnership path (Sedaro et al.)

Part VII — Research: Telemetry Anomaly Detection

Literature Review 01 — State of the Art in Spacecraft Telemetry Anomaly Detection (2023–2026)

Project context: EXHYTE agent — closed-loop Exploration → Hypothesis generation → Testing over spacecraft telemetry.

Compiled: July 2026. All URLs below were visited directly or returned by web search during this review (see Verification note at the end).

Executive summary

The field pivoted hard between 2023 and 2026. The NASA SMAP/MSL benchmarks (Hundman et al., KDD 2018) that dominated the literature for six years are now widely considered broken — trivial anomalies, unrealistic anomaly density, mislabeled ground truth, run-to-failure bias — and the point-adjustment (PA) evaluation protocol used with them provably inflates F1 to the point where random anomaly scores look state-of-the-art. The credible modern reference is the **ESA Anomaly Detection Benchmark (ESA-ADB, 2024)**: ~17.5 years of real telemetry from ESA missions, curated anomaly annotations by spacecraft operations engineers, telecommand streams included, and a hierarchical operator-centric evaluation pipeline. Its headline finding is sobering and useful: a tuned **Telemanom-style LSTM forecaster with nonparametric dynamic thresholding (NDT) plus pruning is still the best-performing algorithm**, classical unsupervised detectors collapse to near-zero event-wise precision on raw telemetry, and time-series foundation models have not yet demonstrated an advantage for anomaly detection anywhere. Meanwhile, 2025–2026 edge-deployment work shows a NAS-compressed forecasting detector retaining ~89% corrected event-wise F0.5 in **59 KB of RAM** — onboard deployment is realistic. For an MVP: start with OPS-SAT-AD for fast iteration, target ESA-ADB Mission1 as the real benchmark, implement forecasting + NDT first, and never report PA-adjusted F1.

1. The ESA Anomaly Detection Benchmark (ESA-ADB)

What it is. A joint Airbus Defence & Space / KP Labs / ESA ESOC product (18-month project under ESA’s AI for Automation roadmap) with three components: (1) the **ESA Anomalies Dataset (ESA-AD)** — real, raw telemetry from three ESA missions (~17.5 years total; the full dataset spans ~224 channels, ~821 telecommand/control signals, ~1,430 annotated events), manually annotated by spacecraft operations engineers; (2) an **evaluation pipeline** with time-aware, operator-centric metrics; (3) **benchmarking results** for classical and deep detectors ([arXiv:2406.17826](https://arxiv.org/abs/2406.17826), [GitHub](https://github.com)).

What the benchmark subset contains. Two of the three missions (Mission3 was dropped for having trivial anomalies and too many gaps). Mission1: 76 channels (58 target/monitored), 18 channel groups, 4 subsystems. Mission2: 100 channels (47 target), 29 groups, 5 subsystems. Each mission exceeds **700 million data points** (>7 GB compressed total). Anomaly density is realistic: 0.57% (Mission2) to 1.80% (Mission1) of annotated points. There are **844 annotated events** across the two missions (anomalies, *rare nominal events*, and communication gaps); ~148 are true anomalies. Hundreds of distinct telecommands with millions of executions are included, prioritized 0–3; discovering which telecommands matter is deliberately part of the challenge. Channels/telecommands are anonymized, which forecloses physics-informed name matching and forces universal data-driven approaches.

Evaluation protocol. A *hierarchical* pipeline reflecting operator priorities: top priority is **no false alarms** (“a high false-positive rate disqualifies an algorithm even if it obtains perfect scores in other aspects” — the paper cites FP rate as the main obstacle to adoption of AD in space operations). Metrics include corrected **event-wise precision/recall/F0.5**, **channel-aware** and **subsystem-aware** scores, **alarming precision** (fragmented detections of one event are penalized), **ADTQC** (timing quality: early/late detection), and **affiliation-based** precision/recall. F0.5 (precision-weighted) is used, not F1.

Benchmark results / current SOTA. *Telemanom-ESA-Pruned* (an ESA-adapted Hundman LSTM forecaster + NDT + pruning) is the clear winner on Mission1 — corrected event-wise F0.5 ≈ 0.79 on the lightweight channel subset (0.968 on anomalies-only), event-wise precision 0.999, and the best alarming precision thanks to NDT merging adjacent detections. Classical unsupervised detectors (PCC, HBOS, iForest, KNN) achieve event-wise F0.5 < 0.001 — drowned in false positives from noise and varying sampling rates. DC-VAE-ESA disappoints on event-wise scores but wins timing/affiliation scores, suggesting better thresholding/postprocessing could rescue reconstruction methods. The authors’ conclusion: “new approaches are necessary to address operators’ needs.” The benchmark remains open — there is an active Kaggle competition ([ESA-ADB challenge](https://kaggle.com/competitions/esa-ADB-challenge)) and a project page listing materials (ai4gs.space-codev.org/esa-ADB); no saturated public leaderboard exists yet as of mid-2026.

Follow-on work (2025–2026). A large-scale deployability study on ESA-ADB compared supervised (Multiscale CNN, GCN, GAT) vs unsupervised (Elliptic Envelope, ECOD) methods and found supervised models stronger overall but identified “a fundamental trade-off between detection capacity and operational efficiency” ([arXiv:2607.07335](https://arxiv.org/abs/2607.07335)). An ECML PKDD 2025 workshop entry built a hierarchical ensemble (shapelet + statistical features, per-channel models, cross-channel aggregation) with leakage-safe time-series CV ([Springer](https://www.springer.com)).

2. NASA SMAP/MSL benchmarks (Hundman lineage): status

Origin. Hundman et al. (KDD 2018) introduced channel-wise LSTM prediction + **nonparametric dynamic thresholding (NDT)** on expert-labeled SMAP (soil-moisture satellite) and MSL (Curiosity rover) telemetry, with encoded telecommand information as model input. Code + data remain available under Apache-2.0 at github.com/khundman/telemanom ([paper](https://arxiv.org/abs/1802.04431), [arXiv:1802.04431](https://arxiv.org/abs/1802.04431)). This is still the single most influential paper in the subfield — ESA-ADB’s best baseline is its direct descendant.

Label-quality criticisms. Wu & Keogh (IEEE TKDE, [arXiv:2009.13807](https://arxiv.org/abs/2009.13807)) showed most popular TSAD benchmarks (including NASA’s) suffer four flaws: **triviality** (one-liners like `diff` of consecutive points achieve SOTA), **unrealistic anomaly density**, **misabeled ground truth**, and **run-to-failure bias** (anomalies clustered at sequence ends). The ESA-ADB paper explicitly calls SMAP/MSL “too trivial” with “inconsistent ground truth” and notes that results published on them “are not indicative.”

Re-evaluation results. Herrmann et al., *CEAS Space Journal* 2024 ([Springer](https://www.springer.com)) re-evaluated deep AD on spacecraft telemetry without point adjustment: several deep methods fell **below a raw-signal baseline**; simple classical methods (GMM, kNN, OCSVM, PCA) consistently beat the Anomaly Transformer; a plain **LSTM was the best performer** across nearly all cases. Papers still report 91–96% F1 on SMAP/MSL (e.g., Sparse Binary Transformers 91.1/91.8; various 2025 hybrids ~95–96) — but these numbers are produced under the inflated PA protocol and should not drive design decisions.

Verdict for us: SMAP/MSL are fine as a *smoke test* (small, ubiquitous, Apache-2.0) but must not be the headline benchmark, and PA-F1 numbers from that literature must not be treated as real SOTA.

3. Best-performing recent methods

- **LSTM forecasting + dynamic thresholding (Telemanom lineage).** Still the strongest published result on ESA-ADB (Telemanom-ESA-Pruned). Its advantage comes as much from **thresholding and postprocessing (NDT, pruning, detection merging)** as from the network itself — ESA-ADB authors highlight this explicitly.
 - **Architecture comparisons.** Lakey & Schlippe (IEEE Aerospace Conf. 2024, [arXiv:2403.12864](https://arxiv.org/abs/2403.12864)) compared CNN/RNN/LSTM/Transformer families on spacecraft telemetry: Transformers help on subtle long-duration anomalies but at heavy compute cost; **LSTMs and CNNs give the best accuracy/efficiency balance**. The 2025 Applied Sciences review ([MDPI](https://www.mdpi.com)) tags **TCNs and GCNs** as promising (precision up to ~94% in surveyed studies) and lists onboard deployment, explainability, and evaluation standardization as the open challenges.
 - **Transformers / hybrids.** A steady 2024–2026 stream (Transformer-LSTM hybrids, temporal-enhanced transformers for irregular-interval telemetry, e.g. [Acta Astronautica](https://www.acta-astronautica.com), [ASR](https://www.asr.org)) reports gains, but almost all on SMAP/MSL-style protocols; independent re-evaluations (Section 2) show transformer AD methods often lose to classical baselines when evaluation is fixed.
 - **GAF+CNN image encodings.** Direct classification of Gramian-Angular-Field images beat forecasting in one IEEE study ([Forecasting vs. Classification, IEEE Xplore](https://arxiv.org/abs/2009.13807)), but on ESA-ADB the edge study below found GAF image classification the *worst* of three approaches (48.6% CEF0.5 vs 92.7% for forecasting+threshold). Treat GAF as a niche option, not a default.
 - **Foundation models (TimesFM, Chronos, MOMENT, Time-MoE, TSPulse).** The evidence is consistently negative-to-neutral for AD: an AAAI-25 workshop study found “traditional statistical and deep learning models often match or outperform TSFMs,” which also “require high computational resources” and don’t improve zero/few-shot AD ([arXiv:2412.19286](https://arxiv.org/abs/2412.19286)); a 2026 study found simple one-liners rival five TSFM families for AD ([OpenReview](https://openreview.net)). The interesting middle path is using a frozen TSFM as a **feature extractor** inside a two-stage detector (ChronosAD, [arXiv:2606.01300](https://arxiv.org/abs/2606.01300)). No published TSFM result leads ESA-ADB.
 - **Ensembles.** Hierarchical per-channel → cross-channel ensembles with leakage-controlled CV are an emerging pragmatic winner on ESA-ADB-style data ([ECML PKDD 2025 workshop](https://www.ecmlpkdd.org)).
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4. Edge / onboard deployment

- **Goetze, Schlippe & Lakey 2026** ([arXiv:2603.29375](https://arxiv.org/abs/2603.29375)): the key result for onboard feasibility. On ESA-ADB, they compared forecasting+NDT (92.7% CEF0.5), direct classification (72.4%), and GAF image classification (48.6%), then ran multi-objective neural-architecture optimization targeting two real platforms: a **CubeSat-class PIC32MZ (200 MHz, 16 MB RAM)** and **OPS-SAT's ARM Cortex-A9 (800 MHz, 1 GB)**. The optimized forecasting model kept **88.8% CEF0.5 at 59 KB RAM** (97% RAM and 99.4% ops reduction) — 0.36% of the CubeSat's memory. Conclusion: a Telemanom-style detector fits comfortably onboard.
 - **Horne et al. 2023**, *Journal of Aerospace Information Systems* ([AIAA](https://aiaa.org)): early demonstration that deep AD can respect CubeSat resource budgets.
 - **Hardware landscape.** A 2024 *Remote Sensing* review of onboard NN inference hardware/software ([MDPI](https://arxiv.org/abs/2504.03891)) covers rad-tolerant FPGAs, Movidius Myriad-class VPUs, and 8-bit quantization practice (e.g., Φ sat-2); FPGA-accelerated quantized CNNs for CubeSats are demonstrated in adjacent domains ([arXiv:2504.03891](https://arxiv.org/abs/2504.03891)). ESA's OPS-SAT flew AD experiments in orbit (KP Labs, [project page](#)); the successor OPS-SAT VOLT continues this line.
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5. Evaluation pitfalls (design constraints for EXHYTE's "Testing" stage)

1. **Point-adjustment (PA) inflation.** Kim et al., AAAI 2022 ([arXiv:2109.05257](https://arxiv.org/abs/2109.05257), [PDF](#)): under PA, *random* anomaly scores achieve SOTA F1 (documented jumps like F1 0.32 → 0.85). Any paper reporting PA-F1 on SMAP/MSL is measuring the protocol, not the model.
 2. **The fixes are partially broken too.** An adversarial stress-test of 12 post-PA metrics ([arXiv:2607.11969](https://arxiv.org/abs/2607.11969), 2026) found **affiliation-F1 gameable on 99% of series** and VUS/ROC-based metrics compromised on ~62–64% of datasets; **PR-based metrics and PA%K resisted best (14–18%)**. Recommendation: prefer PR-based metrics or PA%K; distrust affiliation-F1 and ROC-AUC variants.
 3. **Event-wise vs point-wise.** Operators care about *events* and about precision more than recall. ESA-ADB's corrected event-wise **F0.5** plus **alarming precision** (anti-fragmentation) and **ADTQC** (timing) is the most operator-faithful published protocol; adopt it wholesale rather than inventing our own.
 4. **Low anomaly density.** Realistic density is ~0.6–1.8% of points (ESA-ADB). Detectors tuned on dense-anomaly benchmarks produce unusable FP rates on real telemetry — this is exactly why classical unsupervised methods score <0.001 event-wise precision on Mission1.
 5. **Telecommand-driven false positives / rare nominal events.** The majority of Mission2 annotations are *rare nominal events* — atypical but planned telemetry changes (commanded maneuvers, resets). A detector unaware of telecommands will flag them all. Telecommand conditioning (as in Hundman 2018 and ESA-ADB's design) is mandatory, not optional.
 6. **Information leakage.** Classic TimeEval-style evaluation trains and tests unsupervised detectors on the same full series — unrealistic for online monitoring and unfairly favorable. ESA-ADB enforces proper train/test splits and mission phases; the multivariate-benchmark literature adds that "most anomalies in multivariate benchmarks are effectively univariate" ([arXiv:2606.02670](https://arxiv.org/abs/2606.02670)), so channel-wise models remain competitive.
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6. Annotated source list

#	Source (title — venue/year)	URL
1	European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry — Kotowski et al., arXiv/DMLR, 2024 (rev. 2025)	https://arxiv.org/abs/2406.17826SMARTPANTS_PRESERVED_1958SMARTPANTS_PRESERVED_1959
2	ESA-ADB code repository — GitHub (KP Labs), 2024–	https://github.com/kplabs-pl/ESA-ADBSMARTPANTS_PRESERVED_1972SMARTPANTS_PRESERVED_1973
3	ESA Anomalies Dataset (Zenodo record) — ESA/Airbus/KP Labs, 2024	https://doi.org/10.5281/zenodo.12528696SMARTPANTS_PRESERVED_1986SMARTPANTS_PRESERVED_1987
4	Toward Deployable Satellite Anomaly Detection: A Benchmark Study on Large-Scale ESA-ADB Telemetry — arXiv, 2026	https://arxiv.org/abs/2607.07335SMARTPANTS_PRESERVED_2002SMARTPANTS_PRESERVED_2003
5	Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding — Hundman et al., KDD 2018; telemanom repo	https://github.com/khundman/telemanomSMARTPANTS_PRESERVED_2016SMARTPANTS_PRESERVED_2017
6	Current Time Series Anomaly Detection Benchmarks are Flawed... — Wu & Keogh, IEEE TKDE 2021/2023	https://arxiv.org/abs/2009.13807SMARTPANTS_PRESERVED_2030SMARTPANTS_PRESERVED_2031
7	Unmasking overestimation: a re-evaluation of deep	https://link.springer.com/article/10.1007/s12567-023-00529-5SMARTPANTS_PRESERVED_2044SMARTPANTS_PRESERVED_2045

#	Source (title — venue/year)	URL
	anomaly de- tection in spacecraft telemetry — Herrmann et al., CEAS Space Journal, 2024	
8	Towards a Rigorous Evaluation of Time-series Anomaly Detection — Kim et al., AAAI 2022	https://arxiv.org/abs/2109.05257SMARTPANTS PRESERVED 2058SMARTPANTS PRESERVED 2059
9	Did We Actually Fix It? An Adversarial Stress-Test of Post-PA Metrics — arXiv, 2026	https://arxiv.org/abs/2607.11969SMARTPANTS PRESERVED 2072SMARTPANTS PRESERVED 2073
10	The OPS-SAT benchmark for detecting anomalies in satellite telemetry — Ruszczak, Kotowski, Evans, Nalepa, <i>Scientific Data</i> 12:710, 2025	https://www.nature.com/articles/s41597-025-05035-3SMARTPANTS PRESERVED 2088SMARTPANTS PRESERVED 2089
11	A Comparison of Deep Learning Architectures for Spacecraft Anomaly Detection — Lakey & Schlippe, IEEE Aerospace Conference 2024	https://arxiv.org/abs/2403.12864SMARTPANTS PRESERVED 2104SMARTPANTS PRESERVED 2105
12	Deep Learning- Based Anomaly Detection in Spacecraft Telemetry on Edge Devices — Goetze, Schlippe, Lakey, 2026 (also IEEE SCC 2025 version)	https://arxiv.org/html/2603.29375SMARTPANTS PRESERVED 2118SMARTPANTS PRESERVED 2119

#	Source (title — venue/year)	URL
13	Anomaly Detection in Spacecraft Telemetry: Forecasting vs. Classification — IEEE (Xplore #10794845), 2024	https://ieeexplore.ieee.org/document/10794845/SMARTPANTS_PRESERVED_2132SMARTPANTS_PRESERVED_2133
14	A Review of Anomaly Detection in Spacecraft Telemetry Data — Fejjari et al., <i>Applied Sciences</i> (MDPI), 2025	https://www.mdpi.com/2076-3417/15/10/5653SMARTPANTS_PRESERVED_2148SMARTPANTS_PRESERVED_2149
15	Time Series Foundational Models: Their Role in Anomaly Detection and Prediction — AAAI 2025 Workshop	https://arxiv.org/abs/2412.19286SMARTPANTS_PRESERVED_2162SMARTPANTS_PRESERVED_2163
16	When Foundation Models are One-Liners: Limitations and Future Directions for TSAD — OpenReview, 2026	https://openreview.net/forum?id=H27kvyG4qfSMARTPANTS_PRESERVED_2176SMARTPANTS_PRESERVED_2177
17	ChronosAD: Leveraging Time Series Foundation Models for Accurate Anomaly Detection — arXiv, 2026	https://arxiv.org/pdf/2606.01300SMARTPANTS_PRESERVED_2190SMARTPANTS_PRESERVED_2191

#	Source (title — venue/year)	URL
18	A Hierarchical Ensemble Pipeline for Anomaly Detection in ESA Satellite Telemetry — ECML PKDD 2025 workshops, Springer CCIS 2842	https://link.springer.com/chapter/10.1007/978-3-032-19105-2_7 SMARTPANTS PRESERVED 2204SMARTPANTS PRESERVED 2205
19	Anomaly Detection Using Deep Learning Respecting the Resources on Board a CubeSat — Horne et al., <i>J. Aerospace Information Systems</i> , 2023	https://arc.aiaa.org/doi/10.2514/1.011232 SMARTPANTS PRESERVED 2220SMARTPANTS PRESERVED 2221
20	Review on Hardware Devices and Software Techniques Enabling NN Inference Onboard Satellites — <i>Remote Sensing</i> (MDPI), 2024	https://www.mdpi.com/2072-4292/16/21/3957 SMARTPANTS PRESERVED 2236SMARTPANTS PRESERVED 2237
21	Towards Unbiased Evaluation of Time-series Anomaly Detectors — arXiv, 2024	https://arxiv.org/abs/2409.13053 SMARTPANTS PRESERVED 2250SMARTPANTS PRESERVED 2251
22	Anomalies in Multivariate Time Series Benchmarks Are Mostly Univariate — arXiv, 2026	https://arxiv.org/pdf/2606.02670 SMARTPANTS PRESERVED 2264SMARTPANTS PRESERVED 2265

7. Dataset practicalities

Dataset	Download	Size	Li
ESA-AD / ESA-ADB	https://doi.org/10.5281/zenodo.1252869SMARTPANTS_PRESERVED_2297 (Zenodo, v1.0, June 2024)	11.6 GB compressed (Mission1 3.8 GB, Mission2 4.1 GB, Mission3 3.7 GB); >700M points per mission; plan ≥512 GB disk for full pipeline artifacts	C AI
OPS-SAT-AD (OPSSAT-AD)	https://doi.org/10.5281/zenodo.1258835SMARTPANTS_PRESERVED_2323SMARTPANTS_PRESERVED_2324	Small (2,123 univariate segments, 9 channels; MBs not GBs)	C
NASA SMAP/MSL (telemanom)	https://github.com/khundman/telemanomSMARTPANTS_PRESERVED_2343 (repo links data); Kaggle mirror: https://www.kaggle.com/datasets/patrickfleith/nasa-anomaly-detection-dataset-smap-msl/dataSMARTPANTS_PRESERVED_2345SMARTPANTS_PRESERVED_2346	~100s of MB; 82 channels (55 SMAP + 27 MSL), 105 labeled anomaly sequences	A

8. What this means for our MVP

Recommendation 1 — Dataset strategy: OPS-SAT-AD first, ESA-ADB Mission1 as the real target. Start development against **OPS-SAT-AD** (tiny download, clean train/test split, 30 published baselines to beat, CC BY 4.0) to get the EXHYTE loop running end-to-end in days. Then graduate to **ESA-ADB Mission1, lightweight channel subset (channels 41-46)** as the benchmark we publish against — it is the only benchmark reviewers and ESA/industry partners now trust. Keep SMAP/MSL only as a compatibility smoke test.

Recommendation 2 — First baseline detector: channel-wise forecaster + nonparametric dynamic thresholding (Telemanom lineage). Implement (or wrap the MIT/Apache reference implementations of) an LSTM — or GRU/TCN, which are cheaper — one-step forecaster per target channel, with **NDT thresholding, error smoothing, detection merging, and pruning**. This is literally the current SOTA on ESA-ADB (event-wise F0.5 $\approx$ 0.79; 0.97 anomalies-only) and compresses to 59 KB for onboard use later. The published evidence says thresholding/postprocessing contributes as much as the network — budget engineering time accordingly. A DC-VAE-style reconstruction model is the natural second detector (best timing scores; complements forecasting).

Recommendation 3 — Condition on telecommands from day one. Most Mission2 annotations are *rare nominal events* caused by commanded operations. A detector blind to telecommands will bury operators in false positives — the documented #1 adoption killer. Feed encoded telecommand executions as exogenous inputs (as Hundman did) and, in the EXHYTE Hypothesis stage, make “was this preceded by a telecommand?” the first hypothesis the agent tests.

Recommendation 4 — Adopt the ESA-ADB evaluation pipeline verbatim; ban PA-F1. Reuse the MIT-licensed ESA-ADB metrics code: corrected **event-wise F0.5** (precision-weighted), **alarming precision**, **ADTQC**, channel-aware and subsystem-aware scores. Where a scalar is needed for internal tracking, prefer **PR-based metrics or PA%K** (the only ones that survived the 2026 adversarial stress-test); treat affiliation-F1 and ROC-AUC variants as unreliable. This is also a product feature: EXHYTE’s “Testing” stage can market itself as evaluation-rigorous in a field notorious for inflated claims.

Recommendation 5 — Deprioritize foundation models and GAF encodings for the MVP. Three independent 2025–2026 studies show TSFMs don’t beat simple baselines at AD and cost far more compute; GAF+CNN underperforms forecasting on ESA-ADB by ~44 points of CEF0.5. Revisit TSFMs later in the ChronosAD pattern (frozen TSFM embeddings + light detector head) as an Exploration-stage representation, not as the detector.

Recommendation 6 — Architect for channel-wise + cross-channel aggregation and eventual edge deployment. Evidence says most benchmark anomalies are effectively univariate; per-channel detectors with a hierarchical cross-channel aggregation layer (subsystem-aware) scale to hundreds of channels, map naturally onto EXHYTE’s hypothesis attribution (“which channel/subsystem?”), and are what compresses onto CubeSat-class hardware (59 KB RAM demonstrated). Keep the detector core dependency-light so the same code path can later target quantized/FPGA deployments.

VERIFICATION NOTE

Sources 1–4, 7, 9, 10 (paper), 12, 14, 15, 18 and the Zenodo/GitHub/ai4gs/Kaggle pages were fetched directly during this review (the ESA-ADB paper PDF was downloaded and text-extracted; statistics in Section 1 come from that full text). Sources 5, 6, 8, 11, 13, 16, 17, 19–22 and the OPS-SAT Zenodo DOI were returned by web search and/or confirmed via fetched summaries of citing pages. No URL in this document was fabricated.

Part VIII — Research: LLM Scientific Discovery

LLM Agents for Scientific Discovery & Hypothesis Generation (2024–2026)

Literature review for the EXHYTE agent — a closed-loop AI system (Exploration → Hypothesis generation → Testing) for autonomous scientific reasoning over spacecraft telemetry.

Prepared: 2026-07-19. All sources below were retrieved via web search/fetch during this review; URLs were visited or returned by search (see verification note at end).

1. The EXHYTE framework itself

Our product name maps directly onto a real, PubMed-indexed survey. It is worth adopting its vocabulary because it gives us a defensible, citable process model.

1.1 A PROCESS-CENTRIC SURVEY OF AI FOR SCIENTIFIC DISCOVERY THROUGH THE EXHYTE FRAMEWORK

- **Venue/year:** Research Square preprint (also indexed via PMC), December 17, 2025. DOI: 10.21203/rs.3.rs-8370059/v1.
- **Authors:** Md Musaddaqui Hasib, Sumin Jo, Harsh Sinha, Jifeng Song, Arun Das, Zhentao Liu, Hugh Galloway, Huey Huang, Kexun Zhang, Shou-Jiang Gao, Yu-Chiao Chiu, Lei Li, Yufei Huang.
- **URL:** https://pmc.ncbi.nlm.nih.gov/articles/PMC12776506/SMARTPANTS_PRESERVED_2440 (preprint: https://www.researchsquare.com/article/rs-8370059/v1SMARTPANTS_PRESERVED_2442 ; interactive survey: https://webapps.crc.pitt.edu/exhyte/SMARTPANTS_PRESERVED_2444)
- **The framework, precisely as defined:** EXHYTE formalizes discovery as an iterative cycle of three phases, each with substages:
 - **EXPLORE** — “assemble and structure knowledge so that knowledge gaps become explicit and actionable.” Substages: **E1 Query structuring → E2 Data retrieval → E3 Knowledge assembly.**
 - **HYPOTHESIZE** — “transforms knowledge gaps into ideas or specific, testable claims.” Substages: **H1 Hypothesis/idea generation → H2 Hypothesis/idea prioritization.**
 - **TEST** — “translates hypotheses into executable experiments and closes the feedback loop.” Substages: **T1 Experimental design generation → T2 Iterative refinement.**
- **Maturity assessment (directly load-bearing for our roadmap):** EXPLORE is called the **most mature** stage (strong literature retrieval, RAG pipelines, knowledge-graph construction). **Hypothesis prioritization (H2) “remains underdeveloped.”** The **TEST stage “lags behind,”** with most efforts stuck in simulation / in-silico settings.
- **Closed-loop pathways:** Test results feed back through exactly three routes — **refinement** (adjust parameters), **pivoting** (explore an alternative mechanism), or **sunset** (conclude the investigation). The paper argues the most effective systems “model the epistemic structure of inquiry” while preserving “transparency for human oversight,” with human involvement spanning goal-setting, expert evaluation at prioritization, and validation.

Implication: our telemetry agent is squarely a TEST-and-loop problem — the *least* mature part of the field, which is both the risk and the opportunity. We should reuse EXHYTE’s substage decomposition as our internal state machine (E1→E2→E3→H1→H2→T1→T2) and its three feedback routes (refine / pivot / sunset) as explicit agent transitions.

2. Closed-loop AI-scientist systems: what works, what’s criticized

2.1 COSCIENTIST — AUTONOMOUS CHEMICAL RESEARCH WITH LARGE LANGUAGE MODELS

- **Venue/year:** *Nature* 624, December 2023. Gomes lab (CMU): Boiko, MacKnight, Gomes.
- **URL:** https://www.nature.com/articles/s41586-023-06792-0SMARTPANTS_PRESERVED_2513 (open version: https://pmc.ncbi.nlm.nih.gov/articles/PMC10733136/SMARTPANTS_PRESERVED_2515)
- **Findings:** A GPT-4-driven **modular multi-LLM agent** that plans, designs, and executes experiments using tools — internet search, documentation retrieval, code execution, and robotic/liquid-handler APIs. It optimized real palladium-catalysed cross-couplings semi-autonomously. Key architectural lesson: capability came from **tool-augmentation and hardware-constrained planning** (it plans experiments it can actually run), not from the raw LLM. This is the template for grounding a reasoning agent in an executable action space.

2.2 GOOGLE/DEEPMIND “AI CO-SCIENTIST”

- **Venue/year:** Research blog Feb 2025; published in *Nature*, May 2026. Built on Gemini.
- **URL:** https://research.google/blog/accelerating-scientific-breakthroughs-with-an-ai-co-scientist/SMARTPANTS_PRESERVED_2539 (and https://deepmind.google/blog/co-scientist-a-multi-agent-ai-partner-to-accelerate-research/SMARTPANTS_PRESERVED_2541)
- **Findings:** A **multi-agent** system that generates, debates, ranks, and evolves hypotheses against literature and structured databases. Named agents: **Generation** (proposes hypotheses grounded in literature), **Reflection** (a virtual peer reviewer critiquing correctness/novelty/rigour), **Ranking** (an Elo-scored “idea tournament” of pairwise simulated debates), and **Evolution** (refines and recombines top ideas). Domain experts rated its outputs higher in novelty across 15 biomedical goals; wet-lab trials confirmed drug-repurposing (AML) and liver-fibrosis targets. **The single most transferable pattern: separate generation from critique from ranking, and make ranking a tournament with an explicit score.** This is our answer to EXHYTE’s underdeveloped H2 (prioritization).

2.3 ROBIN (FUTUREHOUSE) — END-TO-END MULTI-AGENT DISCOVERY

- **Venue/year:** FutureHouse research announcement, 2025.
- **URL:** https://www.futurehouse.org/research-announcements/demonstrating-end-to-end-scientific-discovery-with-robin-a-multi-agent-systemSMARTPANTS_PRESERVED_2571SMARTPANTS_PRESERVED_2572
- **Findings:** Automates the full loop — literature review → hypothesis → experimental design → data analysis → iterative refinement — via three specialized agents: **Crow** (targeted lit search via PaperQA2), **Falcon** (deep literature review/evaluation), and **Finch** (autonomous data analysis across experimental modalities). It identified ripasudil as a novel candidate for dry AMD. Demonstrates that **role-specialized agents wired around a shared literature/analysis substrate** can close a loop end-to-end in a real domain.

2.4 PAPERQA2 / FUTUREHOUSE PLATFORM (THE EXPLORE SUBSTRATE)

- **Venue/year:** FutureHouse, 2024–2025 (PaperQA2 and ether0 open-source; Edison Scientific spun out Nov 2025).
- **URL:** https://www.futurehouse.org/aboutSMARTPANTS_PRESERVED_2597 (guide: https://intuitionlabs.ai/articles/futurehouse-ai-agents-platformSMARTPANTS_PRESERVED_2599)
- **Findings:** PaperQA2 was the **first AI agent reported to beat PhD-level researchers head-to-head on scientific literature tasks** (retrieval, summarization, contradiction detection). ether0 is a 24B open-weights chemistry reasoning model outperforming general LLMs on molecular design. Confirms EXHYTE’s claim that the EXPLORE stage is mature — we should **buy/reuse RAG-over-literature rather than build it**, and spend our effort on TEST.

2.5 SAKANA “AI SCIENTIST” — THE CAUTIONARY TALE

- **Venue/year:** Sakana AI, 2024; independent evaluation: “Evaluating Sakana’s AI Scientist: Bold Claims, Mixed Results, and a Promising Future?”, arXiv:2502.14297 (2025), later in *ACM SIGIR Forum*.
- **URL:** https://arxiv.org/abs/2502.14297SMARTPANTS_PRESERVED_2623SMARTPANTS_PRESERVED_2624
- **Findings / criticisms (read these as our failure catalog):** literature review relied on **shallow keyword search**, producing **poor novelty judgments** — well-known ideas (e.g., micro-batching for SGD) were flagged as novel. **42% of experiments failed on coding errors**; others produced flawed/misleading results. Manuscripts had **hallucinated numerical results**, missing figures, repeated sections, placeholder text. Verdict: it lacks scientific reasoning and critical self-assessment. Upside: a full paper cost **\$6-\$15 and ~3.5 h of human time**. The lesson for us: **cheap generation is real, but unverified generation is worthless — every claim must be tied to an executable check.**

3. LLM hypothesis-generation quality: benchmarks & evaluation

3.1 HYPOBENCH — SYSTEMATIC BENCHMARKING OF HYPOTHESIS GENERATION

- **Venue/year:** arXiv:2504.11524 (2025), ChicagoHAI.
- **URL:** https://arxiv.org/abs/2504.11524SMARTPANTS_PRESERVED_2656 (site: https://chicagohai.github.io/HypoBench/SMARTPANTS_PRESERVED_2658)
- **Findings:** 7 real-world + 5 synthetic tasks, 194 datasets, with **controlled difficulty knobs — noise, distractors, compositional depth** — to test robustness. Evaluates practical utility, generalizability, and **hypothesis discovery rate** (does the method actually recover the ground-truth explanatory structure). This is the right frame for us: don’t just score “does it sound plausible,” score **“does it recover the true generative mechanism under noise”** — exactly the telemetry setting.

3.2 LIVEIDEABENCH — CREATIVITY/DIVERGENT THINKING WITH MINIMAL CONTEXT

- **Venue/year:** [arXiv:2412.17596 \(Dec 2024\)](#).
- **URL:** https://arxiv.org/abs/2412.17596SMARTPANTS_PRESERVED_2682SMARTPANTS_PRESERVED_2683
- **Findings:** Evaluates scientific idea generation from single-keyword prompts using a dynamic panel of LLM judges, scoring **originality, feasibility, fluency, flexibility** (Guilford's creativity dimensions). Useful signal: **feasibility and novelty trade off**, and free-form, low-context prompting maximizes divergence but not testability. Argues for constraining prompts when you need actionable, not merely creative, hypotheses.

3.3 TRUTHHYPO — TRUTHFULNESS, NOT JUST NOVELTY

- **Venue/year:** "Toward Reliable Scientific Hypothesis Generation: Evaluating Truthfulness and...", IJCAI 2025.
- **URL:** https://www.ijcai.org/proceedings/2025/0873.pdfSMARTPANTS_PRESERVED_2704SMARTPANTS_PRESERVED_2705
- **Findings:** Argues most prior work over-indexes on **novelty/diversity and ignores truthfulness**. TruthHypo integrates **literature retrieval + knowledge-graph retrieval** to evaluate whether generated hypotheses are actually grounded/true, not just surprising. Directly relevant: a novel-sounding failure mechanism that contradicts the physics of the subsystem is worse than useless. We need a truthfulness gate backed by a domain knowledge graph (subsystem topology, known failure modes).

3.4 AI IDEA BENCH 2025

- **Venue/year:** [arXiv:2504.14191 \(2025\)](#).
- **URL:** https://arxiv.org/pdf/2504.14191SMARTPANTS_PRESERVED_2726SMARTPANTS_PRESERVED_2727
- **Findings:** A research-idea-generation benchmark comparing LLM-generated ideas against actual accepted-paper ideas, combining quantitative reference-based metrics with LLM-judge assessment of novelty/feasibility/quality. Reinforces the emerging consensus that **LLM-judge panels plus reference grounding** are the practical evaluation harness — useful for our internal "is this hypothesis good?" scorer.

4. LLM agents for root-cause analysis / diagnostics (closest commercial analog)

This is the most transferable literature: telemetry RCA in AIOps and industrial fault diagnosis is structurally identical to our TEST-and-diagnose loop over spacecraft sensors.

4.1 EXPLORING LLM-BASED AGENTS FOR ROOT CAUSE ANALYSIS

- **Venue/year:** [arXiv:2403.04123 \(2024\)](#), Microsoft-affiliated.
- **URL:** https://arxiv.org/abs/2403.04123SMARTPANTS_PRESERVED_2751SMARTPANTS_PRESERVED_2752
- **Findings:** Studies LLM agents (e.g., ReAct-style, with tools to collect diagnostic data) for cloud-incident RCA; references RCACopilot-style pipelines that use **predefined handlers to gather multi-modal diagnostic data**, then an LLM to predict a root-cause category. Key lesson: **RCA quality is dominated by retrieval/tooling that pulls the right evidence, not by the LLM's free reasoning**. Agents that can call diagnostic collectors beat pure prompting.

4.2 AIOPSLAB — HOLISTIC EVALUATION FRAMEWORK FOR AUTONOMOUS-CLOUD AGENTS

- **Venue/year:** [arXiv:2501.06706 \(2025\)](#), Microsoft Research; also FSE 2025.
- **URL:** https://arxiv.org/abs/2501.06706SMARTPANTS_PRESERVED_2773 (blog: https://www.microsoft.com/en-us/research/blog/aiopslab-building-ai-agents-for-autonomous-clouds/SMARTPANTS_PRESERVED_2775)
- **Findings:** An open-source framework that deploys microservice environments, **injects faults, generates workloads, and exports telemetry**, then scores agents on detection → localization → root-cause diagnosis → mitigation. This is essentially the **eval harness we should copy**: fault injection + telemetry export + staged scoring gives us a repeatable way to measure whether our agent's hypotheses are correct against known ground truth.

4.3 ITBENCH — SRE/KUBERNETES INCIDENT RCA BENCHMARK

- **Venue/year:** IBM Research, 2025 (leaderboard on Artificial Analysis).
- **URL:** https://artificialanalysis.ai/evaluations/itbench-aaSMARTPANTS_PRESERVED_2797SMARTPANTS_PRESERVED_2798
- **Findings:** Benchmarks agents on real-world SRE tasks (Kubernetes incident root-cause) using standard tool-use interaction; a **dynamic-environment** benchmark. Confirms the field is converging on **agent-in-a-simulated-ops-environment** as the evaluation standard, and that current agents are far from solving even scoped RCA — set realistic accuracy expectations for our MVP.

4.4 LLM-TSFD — INDUSTRIAL TIME-SERIES FAULT DIAGNOSIS, HUMAN-IN-THE-LOOP

- **Venue/year:** [Expert Systems with Applications, Vol. 264, 2024.](#)
- **URL:** https://www.sciencedirect.com/science/article/abs/pii/S0957417424027283SMARTPANTS_PRESERVED_2821SMARTPANTS_PR
- **Findings:** A **task-driven, human-in-the-loop** LLM method for industrial time-series fault diagnosis that reduces reliance on manually annotated data, improves interpretability, and keeps a human in the correction loop. Directly validates our intended posture: **the LLM proposes and explains; the expert confirms/corrects; the loop learns.** Not full autonomy.

4.5 EVALUATING LLMs FOR TIME-SERIES ANOMALY DETECTION IN AEROSPACE SOFTWARE

- **Venue/year:** [arXiv:2601.12448 \(2026\).](#)
- **URL:** https://arxiv.org/pdf/2601.12448SMARTPANTS_PRESERVED_2843SMARTPANTS_PRESERVED_2844
- **Findings (the sobering, most MVP-relevant result):** In aerospace time-series settings, **LLMs do not consistently beat established statistical/ML baselines.** Documented failure modes: handling **long continuous sequences**, maintaining **numerical precision** for thresholds, **generalizing across sensor types/configurations**, and compute cost vs lightweight detectors. Recommendation: **hybrid architectures** (statistical detectors for detection; LLM for reasoning/explanation), extensive domain validation, and always benchmarking against statistical baselines. **Do not feed raw numeric telemetry to the LLM and ask “is this anomalous.”**

4.6 DML-LLM HYBRID FOR FAULT DETECTION IN SENSOR-RICH SYSTEMS

- **Venue/year:** [PMC \(2025\). “DML-LLM Hybrid Architecture for Fault Detection and Diagnosis...”](#)
- **URL:** https://pmc.ncbi.nlm.nih.gov/articles/PMC13030379/SMARTPANTS_PRESERVED_2873SMARTPANTS_PRESERVED_2874
- **Findings:** A hybrid of deep ML (detection) + LLM (diagnosis/explanation) cut detection time from **7.4 h → 1.2 h** and improved F1 from **0.59 → 0.83**, giving scalable, explainable diagnosis. Concrete evidence that the **detector-does-numbers / LLM-does-narrative** split is the winning pattern in sensor-rich industrial settings — our telemetry analog.

5. Human-AI collaboration & uncertainty communication for expert review

5.1 HUMAN-AI COLLABORATIVE UNCERTAINTY QUANTIFICATION

- **Venue/year:** [arXiv:2510.23476 \(2025\), also OpenReview.](#)
- **URL:** https://arxiv.org/abs/2510.23476SMARTPANTS_PRESERVED_2900SMARTPANTS_PRESERVED_2901
- **Findings:** Formalizes an **AI refining a human expert’s proposed prediction set** with two explicit goals: **avoid counterfactual harm** (don’t degrade the human’s correct judgments) and **complementarity** (recover outcomes the human missed). Frames the review UX we want: the agent should present a **set of candidate hypotheses** with calibrated confidence and let the expert prune — not hand down a single answer.

5.2 NOT ALL UNCERTAINTY IS EQUAL: UNCERTAINTY GRANULARITY IN LLM-ASSISTED DECISIONS

- **Venue/year:** [arXiv:2605.28571 \(2026\).](#)
- **URL:** https://arxiv.org/pdf/2605.28571SMARTPANTS_PRESERVED_2926SMARTPANTS_PRESERVED_2927
- **Findings:** Shows the **granularity** of communicated uncertainty (per-claim vs whole-answer) materially changes how effectively humans verify LLM outputs. Fine-grained, per-claim uncertainty helps experts focus verification effort; coarse “overall confidence” scores are largely ignored or cause automation bias. Design cue: attach uncertainty **to each mechanistic step of a hypothesis**, not to the hypothesis as a whole.

5.3 TRUST CALIBRATION & STRENGTH/WEAKNESS TRANSPARENCY (SUPPORTING)

- **Venue/year:** [Frontiers in Computer Science, 2025 \(“Why not both? Complementing explanations with uncertainty and self-confidence”\); Springer, 2025 \(“Beyond Predictions: AI Strength and Weakness Transparency”\).](#)
- **URL:** https://www.frontiersin.org/journals/computer-science/articles/10.3389/fcomp.2025.1560448/fullSMARTPANTS_PRESERVED_2948SMARTPANTS_PRESERVED_2949
- **Findings:** Combining **explanations + calibrated uncertainty + disclosed self-confidence** improves trust calibration and reduces automation bias; conveying where the AI is weak (not just its answer) measurably improves joint performance. For us: the agent should openly flag “low evidence / out-of-distribution subsystem” states rather than always producing a confident hypothesis.

What this means for our MVP

Architecture patterns to adopt

1. **Model the loop on EXHYTE substages as an explicit state machine** — E1 query-structuring → E2 retrieval → E3 knowledge-assembly → H1 generate → H2 prioritize → T1 design-test → T2 refine — with the three defined feedback routes as first-class transitions: **refine / pivot / sunset**. This gives us a citable, legible control flow instead of an opaque agent loop.
2. **Separate roles into distinct agents (co-scientist / Robin pattern)**: a Generator, a Reflection/critic (“virtual reviewer” for correctness + physical plausibility), a Ranker, and an Evolver. **Make prioritization (H2) a scored tournament** (Elo-style pairwise debate) — this is the field’s best current answer to the substage EXHYTE flags as underdeveloped, and prioritization is where expert time is most leveraged.
3. **Hybrid detector + LLM, never LLM-on-raw-numbers** (aerospace anomaly + DML-LLM evidence). Lightweight statistical/ML detectors own anomaly *detection* and numeric thresholds; the LLM owns *diagnosis, mechanism, and explanation* over structured, pre-digested features. Expected payoff mirrors the industrial result (F1 0.59→0.83, 7.4h→1.2h).
4. **Tool-grounded reasoning (Coscientist pattern)**: the agent must act through tools — telemetry queries, subsystem-model lookups, code execution, simulation — and should only plan tests it can actually run against available data/sims. Capability lives in the tools, not the prompt.
5. **Buy the EXPLORE layer**: literature/knowledge retrieval and RAG are mature (PaperQA2 beats PhDs). Reuse an off-the-shelf RAG/KG stack for E1-E3 and spend engineering on TEST (T1-T2), the field’s weakest link and our differentiator.
6. **Human-in-the-loop by design**: present a *ranked set* of hypotheses with **per-mechanistic-step calibrated uncertainty** (not a single confidence number), let experts prune/correct, and feed corrections back — optimizing for complementarity and against counterfactual harm.

Pitfalls to avoid

- **Hallucinated mechanisms / fabricated numbers** (Sakana: hallucinated results, 42% experiment failures; TruthHypo: novelty without truth). Every proposed mechanism must be **grounded against a domain knowledge graph** (subsystem topology + known failure modes) and rejected if it contradicts known physics. Add a truthfulness gate, not just a novelty score.
- **Shallow retrieval → false novelty** (Sakana). Do not equate “not in my keyword search” with “novel.” Prioritization must check against a real failure-mode KB.
- **Feeding raw numeric time-series to the LLM** — it underperforms simple statistical baselines and fails on long sequences and numeric precision. Pre-digest.
- **Over-trusting a single confident answer** — coarse whole-answer confidence invites automation bias; fine-grained per-claim uncertainty is what actually helps experts verify.
- **Aiming for full autonomy in TEST** — the field’s TEST stage is immature and mostly in-silico; scope the MVP to human-confirmed loops (LLM-TSFD posture).

How to make hypotheses machine-testable

- **Emit hypotheses as structured, executable objects, not prose**. Each hypothesis = { candidate mechanism; affected subsystem/signals; a *predicted, checkable consequence* (e.g., “if thermal-latch degradation, expect signal X to lead signal Y by N samples under load Z”); the concrete test/query/sim to run; a pass/fail predicate; prior evidence links }. This is EXHYTE’s own definition of a good hypothesis (“specific, testable claims”) operationalized.
- **Bind every claim to an executable check** (Coscientist/AIOpsLab lesson) — a telemetry query, statistical test, or simulation the agent can actually run and whose result routes it to refine/pivot/sunset.
- **Score against ground truth with a fault-injection eval harness** modeled on AIOpsLab/ITBench and HypoBench: inject known faults into telemetry (or replay labeled anomalies), and measure **hypothesis discovery rate under noise/distractors** — did the agent recover the true generative cause, not just a plausible story.
- **Prefer constrained/templated generation over free-form** when testability matters (LiveldeaBench: free-form maximizes divergence, not actionability). The structured-object template above is the constraint.

VERIFICATION NOTE

All 19 sources cited were returned by web search and/or fetched during this review. The EXHYTE paper (PMC12776506), Coscientist (Nature 2023), Sakana evaluation (arXiv:2502.14297), Google co-scientist, Robin/FutureHouse, HypoBench (2504.11524), LiveldeaBench (2412.17596), TruthHypo (IJCAI 2025), AI Idea Bench (2504.14191), LLM-RCA (2403.04123),

[AIOpsLab \(2501.06706\)](#), [ITBench](#), [LLM-TSFD](#), [the aerospace anomaly paper \(2601.12448\)](#), [DML-LLM hybrid \(PMC13030379\)](#), and the three human-AI/uncertainty papers all correspond to URLs surfaced by search. The EXHYTE PMC page and the aerospace anomaly PDF were fetched and their content read directly. No citations were fabricated.

Part IX — Research: Digital Twins & SBI

Literature Review 03 — Digital Twins for Spacecraft, Simulation-Based Inference, and Experiment Design (2022–2026)

Scope: Sources supporting the Testing layer of the EXHYTE agent: an open-source digital twin whose parameters can encode fault hypotheses, simulation ensembles compared against real telemetry via Bayesian model comparison, and Bayesian experimental design for next-observation planning.

Date: 2026-07-19. All URLs below were visited directly or returned by web search during this review.

1. Open-source spacecraft simulators usable as a digital twin

1.1 BASILISK (AVS LAB, UNIVERSITY OF COLORADO / LASP)

- **Year:** actively maintained; current release 2.11.0 (2026). Framework paper 2020.
- **URL:** https://avslab.github.io/basilisk/SMARTPANTS_PRESERVED_3091 | Framework paper: https://arc.aiaa.org/doi/10.2514/1.1010762SMARTPANTS_PRESERVED_3093SMARTPANTS_PRESERVED_3094
- C/C++ modules wrapped for Python via SWIG under the permissive **ISC license** — every simulation is a Python script, which makes programmatic parameter sweeps trivial. Targets **faster-than-realtime** execution (stated goal: a mission year of 6-DOF dynamics at $\geq 365\times$ speed-up) and supports **bit-for-bit repeatable Monte Carlo runs** with parameter dispersions drawn from configurable distributions — exactly the ensemble machinery a hypothesis-testing loop needs.
- Subsystem coverage: full attitude dynamics and ADCS (reaction wheels incl. friction/jitter models, thrusters, coarse sun sensors, star trackers, magnetometers/torque bars), flexible/multi-body dynamics, fuel slosh, and a **power/EPS subsystem** ([simpleBattery](#) , [simpleSolarPanel](#) with eclipse/attitude coupling, [simplePowerSink](#) ; see https://hanspeterschaub.info/basilisk/Documentation/simulation/power/index.htmlSMARTPANTS_PRESERVED_3107). Thermal modeling is comparatively thin (device-level, no full thermal network).
- A 2025 preprint bridges Basilisk to ROS 2 for modular sim and hardware integration, evidence of an active integration ecosystem: https://arxiv.org/pdf/2512.09833SMARTPANTS_PRESERVED_3111SMARTPANTS_PRESERVED_3112

1.2 NASA 42 (GODDARD)

- **Year:** open source since 2014, still updated. NTRS overview paper 2018.
- **URL:** https://github.com/ericstoneking/42SMARTPANTS_PRESERVED_3125 | https://ntrs.nasa.gov/citations/20180000954SMARTPANTS_PRESERVED_3127 | Catalog: https://software.nasa.gov/software/GSC-16720-1SMARTPANTS_PRESERVED_3129SMARTPANTS_PRESERVED_3130
- General-purpose simulation of attitude and orbit dynamics of multiple spacecraft with rigid or flexible multi-body structures; written in C under the **NASA Open Source Agreement (NOSA)**. Rigorous enough to support the full spacecraft lifecycle from concept studies through I&T, and models two- and three-body orbital regimes anywhere in the solar system.
- No native Python bindings; integration is via a **socket-based IPC interface**, so driving 42 from a Python inference loop means managing input decks and sockets rather than calling functions. NOSA is also not OSI-approved, which matters for a commercial product embedding it.

1.3 NASA NOS3 — OPERATIONAL SIMULATOR FOR SPACE SYSTEMS (CFS + 42 + HARDWARE MODELS)

- **Year:** open-sourced ~2019; actively developed; Design Reference Mission paper 2023.
- **URL:** https://github.com/nasa/nos3SMARTPANTS_PRESERVED_3151 | DRM paper: https://ntrs.nasa.gov/citations/20230010613SMARTPANTS_PRESERVED_3153SMARTPANTS_PRESERVED_3154
- NASA IV&V's software-only testbed that stitches together **cFS flight software, the 42 dynamics engine, software models of COTS hardware (EPS, reaction wheels, GPS, etc.), and COSMOS ground software** into one virtual spacecraft; NASA explicitly describes it as an “open-source digital twin” for smallsat development, V&V, and operations. License is NOSA 1.3.
- This is the closest open-source analog to a full closed-loop twin (flight software in the loop, command and telemetry paths included), but it runs as an Ubuntu VM of many cooperating processes — heavyweight for running thousands of ensemble members, and better suited as a later high-fidelity validation stage than as the MVP inference simulator.

1.4 OREKIT

- **Year:** mature, actively maintained (CS GROUP).

- **URL:** https://github.com/CS-SI/OrekitSMARTPANTS_PRESERVED_3173 | Python bindings: https://anaconda.org/conda-forge/orekit_jpyeSMARTPANTS_PRESERVED_3175SMARTPANTS_PRESERVED_3176
- Low-level space flight dynamics library in Java under **Apache 2.0**: orbits, frames, propagation (numerical and semi-analytical), attitude pointing laws, event detection, and operational-grade **orbit determination**. Two Python routes exist (JCC-based wrapper and the newer `orekit_jpye`).
- Strong for precision orbit/OD problems (e.g., drag or maneuver-anomaly hypotheses) but it is not a spacecraft subsystem simulator — no reaction wheel, EPS, or thermal device models — so it complements rather than replaces an ADCS-centric twin.

1.5 POLIASTRO (ARCHIVED) → HAPSIRA (FORK)

- **Year:** poliastro archived October 2023; hapsira fork 0.18.x maintained since.
 - **URL:** https://github.com/poliastro/poliastroSMARTPANTS_PRESERVED_3197 | https://github.com/pleiszenburg/hapsiraSMARTPANTS_PRESERVED_3199SMARTPANTS_PRESERVED_3200
 - Pure-Python, MIT-licensed interactive astrodynamics (two-body orbit propagation, Lambert solvers, plotting, units handling). poliastro itself is **archived and unmaintained**; hapsira continues development with an emphasis on speed, but with a single maintainer.
 - Covers orbital mechanics only — no attitude dynamics, actuators, or subsystem models — so it is unsuitable as the fault-hypothesis twin; at most a convenience library for orbit geometry utilities.
-

2. Spacecraft digital twin literature and programs

2.1 ORION SYSML MODEL, DIGITAL TWIN, AND LESSONS LEARNED FOR ARTEMIS I (NASA)

- **Year:** 2022 (NTRS), presented 2023.
- **URL:** https://ntrs.nasa.gov/citations/20220017217SMARTPANTS_PRESERVED_3222SMARTPANTS_PRESERVED_3223
- NASA's Orion digital twin pilot (started 2020) built an executable SysML model connecting design data to unit-specific physical asset information to support Artemis I flight operations. The stated payoff is directly relevant to EXHYTE's pitch: digital modeling can "reduce the time to answer questions by days and required human resources by an order of magnitude" during anomaly response.
- Notably, this is a *systems-engineering* twin (MagicDraw/Cameo), not a physics-simulation twin — evidence that even NASA's flagship twin effort focuses on structured knowledge plus telemetry rather than closed-loop simulation, leaving the physics-ensemble niche open.

2.2 DIGITAL TWIN OF SPACE ENVIRONMENT: DEVELOPMENT, CHALLENGES, APPLICATIONS, AND FUTURE OUTLOOK

- **Year:** 2024 (Remote Sensing 16(16):3023).
- **URL:** https://www.mdpi.com/2072-4292/16/16/3023SMARTPANTS_PRESERVED_3242SMARTPANTS_PRESERVED_3243
- Survey of digital twin work across the space domain: spacecraft health management, astronaut training, station maintenance, reusable rockets, and ESA/NASA program activity (including ESA's on-orbit-servicing twins and Digital Twin Earth). Useful as the state-of-the-art map and for its catalog of open challenges — model fidelity, real-to-sim data assimilation, and validation — that align with the gaps EXHYTE targets.

2.3 DIGITAL TWIN CONSTRUCTION FROM FAULT AND BEHAVIOUR TREES (MODELS-C 2022)

- **Year:** 2022 (ACM MODELS Companion).
- **URL:** https://dl.acm.org/doi/10.1145/3550356.3561550SMARTPANTS_PRESERVED_3258SMARTPANTS_PRESERVED_3259
- Proposes building digital twin models for spacecraft *operations* directly from fault trees and behaviour trees, i.e., encoding failure knowledge as executable twin structure. Relevant precedent for EXHYTE's Hypothesis→Testing mapping: fault hypotheses can be represented as structured model variants rather than free-form parameter changes.

2.4 SEDARO (COMMERCIAL)

- **Year:** company active; Sedaro 4.0 release 2024–2025 era.
- **URL:** https://sedaro.com/SMARTPANTS_PRESERVED_3276 | Docs/API: https://docs.sedaro.com/SMARTPANTS_PRESERVED_3278SMARTPANTS_PRESERVED_3279

- Cloud-native SaaS for space-system digital twins with agent-based, high-fidelity mission simulation; exposes **OpenAPI clients (Python, MATLAB, etc.)** and advertises use cases including diagnosing and responding to on-orbit faults, operator training, and testing commands/software updates against the twin. SpaceNews reports digital twins gaining traction in military satellite programs with Sedaro among the named vendors (https://spacenews.com/digital-twins-gaining-traction-in-military-satellite-programs/SMARTPANTS_PRESERVED_3284).
- Competitive signal: the *twin platform* itself is being commoditized; EXHYTE's differentiation must be the autonomous reasoning loop on top, not the twin.

2.5 ANTARIS TRUETWIN (COMMERCIAL)

- **Year:** active; Aalyria Spacetime integration announced 2024.
- **URL:** https://www.antis.space/truetwinSMARTPANTS_PRESERVED_3302SMARTPANTS_PRESERVED_3303
- "Full Mission Virtualization": TrueTwin composes flight software (SatOS), subsystem software, and hardware models into a virtual satellite that processes real operational commands and generates real responses, before hardware exists — including simultaneous multi-mission constellation simulation. Another confirmation that command-in/telemetry-out virtual satellites are commercially established, while *closed-loop hypothesis testing against live telemetry* remains the open layer.

3. Simulation-based inference (SBI)

3.1 THE `sbi` PACKAGE (SBI-DEV)

- **Year:** ongoing; v0.26.1 released April 2026; "sbi reloaded" paper 2024.
- **URL:** https://github.com/sbi-dev/sbiSMARTPANTS_PRESERVED_3323 | Paper: https://arxiv.org/pdf/2411.17337SMARTPANTS_PRESERVED_3325SMARTPANTS_PRESERVED_3326
- PyTorch toolkit under **Apache 2.0** implementing Neural Posterior Estimation (NPE-A/B/C, TSNPE, FMPE, NPSE), Neural Likelihood Estimation (NLE, MNLE), Neural Ratio Estimation (NRE-A/B/C, BNRE), plus classical ABC, diagnostics (e.g., simulation-based calibration), and embedding networks for learned summary statistics. Distinguishes **amortized** methods (train once on simulations, then infer near-instantly for any new observation) from sequential per-observation methods.
- Fits EXHYTE directly: the Basilisk twin is the simulator, fault parameters (friction coefficient, sensor bias/drift rate, panel degradation) are θ , and real telemetry windows are x ; amortized NPE gives posteriors over fault parameters in milliseconds at operations time.

3.2 FAST BAYESIAN EQUIPMENT CONDITION MONITORING VIA SBI (HEAT EXCHANGER HEALTH)

- **Year:** 2026 (arXiv, April).
- **URL:** https://arxiv.org/abs/2604.20735SMARTPANTS_PRESERVED_3347SMARTPANTS_PRESERVED_3348
- Closest published analog of EXHYTE's Testing layer on an engineered system: amortized NPE trained on a physics simulator diagnoses fouling and leakage failure modes from thermal-fluid observations, recovering full posteriors over degradation parameters. Achieves diagnostic accuracy comparable to MCMC with an **82x inference speedup**, by front-loading simulation cost into offline training — the same architecture (offline ensemble training → online near-instant hypothesis scoring) EXHYTE needs for telemetry-rate operation.

3.3 BAYESFLOW — AMORTIZED BAYESIAN WORKFLOWS AND MODEL COMPARISON

- **Year:** 2023 (IOSS paper); actively developed.
- **URL:** https://arxiv.org/abs/2306.16015SMARTPANTS_PRESERVED_3365 | https://bayesflow.org/stable-legacy/index.htmlSMARTPANTS_PRESERVED_3367SMARTPANTS_PRESERVED_3368
- Python library for amortized inference with generative networks that, unlike `sbi`, ships first-class **amortized Bayesian model comparison**: train a neural classifier over competing simulators to output posterior model probabilities / Bayes factors, or combine amortized posterior+likelihood estimation to compute evidence. This maps one-to-one onto EXHYTE's "which fault hypothesis best explains the telemetry" question, where each hypothesis is a differently-parameterized (or structurally different) twin configuration.

3.4 SBI WITH NPE APPLIED TO X-RAY SPECTRAL FITTING (A&A)

- **Year:** 2024.
- **URL:** https://www.aanda.org/articles/aa/full_html/2024/06/aa49214-24/aa49214-24.htmlSMARTPANTS_PRESERVED_3385SMARTPANTS_PRESERVED_3386

- [Demonstrates NPE working correctly on real instrument data down to the low-count Poisson regime, validating SBI on noisy, discretized sensor streams structurally similar to spacecraft telemetry channels. Useful methodological reference for handling detector noise models inside the simulator rather than the likelihood.](#)

3.5 SBI FOR STRUCTURAL SYSTEM IDENTIFICATION, MODEL UPDATING, AND DAMAGE DETECTION

- **Year:** 2026 (Engineering with Computers).
- **URL:** <https://link.springer.com/article/10.1007/s00366-026-02332-2>SMARTPANTS PRESERVED 3401SMARTPANTS PRESERVED 3402
- [Applies simulation-based inference to structural health monitoring: identifying model parameters and detecting damage from response measurements when likelihoods are intractable. Together with 3.2, establishes a small but growing precedent for SBI on *engineered* \(not just natural-science\) systems — supporting the claim that SBI transfers to spacecraft sub-system fault parameters.](#)

3.6 MACHINE LEARNING FOR REACTION WHEEL FAULT DETECTION USING SIMULATED TELEMETRY DATA (CU BOULDER, USES BASILISK)

- **Year:** 2022 (AIAA SciTech Forum).
- **URL:** <https://doi.org/10.2514/6.2022-2507>SMARTPANTS PRESERVED 3419 | PDF: <https://hanspeterschaub.info/Papers/Carneiro2022.pdf>SMARTPANTS PRESERVED 3421SMARTPANTS PRESERVED 3422
- [Carneiro, Schaub, Lahijanjan et al. generate synthetic telemetry with **Basilisk**, inject reaction-wheel faults \(including wheel-speed encoder faults\), and train supervised and unsupervised ML detectors on the resulting stream. Direct proof-of-concept that Basilisk telemetry ensembles with injected ADCS faults are realistic enough to train fault-reasoning models — de-risking EXHYTE's choice of Basilisk as the fault-hypothesis twin.](#)

4. Bayesian optimal experimental design and active learning

4.1 PYRO PYRO.CONTRIB.OED — EIG ESTIMATION FRAMEWORK

- **Year:** released 2020 (Uber AI), maintained in Pyro.
- **URL:** <https://docs.pyro.ai/en/dev/contrib.oed.html>SMARTPANTS PRESERVED 3442 | Announcement: <https://www.uber.com/us/en/blog/oed-pyro-release/>SMARTPANTS PRESERVED 3444SMARTPANTS PRESERVED 3445
- [The most practical off-the-shelf BOED toolkit: estimators for **Expected Information Gain** including nested Monte Carlo, Laplace approximation \(`laplace eig` \), variational bounds \(`vi eig` , `vnmc eig` \), and marginal-entropy estimators \(`marginal eig` \), optimizable with standard Pyro optimizers. For EXHYTE v1, EIG over a *discrete* menu of candidate observations \(which telemetry channels to downlink at high rate, which diagnostic maneuver/actuation test to run\) needs only \[these estimators plus the already-built simulator\]\(#\).](#)

4.2 DEEP ADAPTIVE DESIGN (DAD) — AMORTIZED SEQUENTIAL BOED

- **Year:** 2021 (ICML).
- **URL:** <https://proceedings.mlr.press/v139/foster21a.html>SMARTPANTS PRESERVED 3464 | Code: <https://github.com/ae-foster/dad>SMARTPANTS PRESERVED 3466SMARTPANTS PRESERVED 3467
- [Foster, Ivanova et al. train a design policy network offline so that at deployment the next experiment is chosen in **milliseconds via a single forward pass**, conditioned on all previous designs and outcomes — removing the per-step EIG optimization that makes classical sequential BOED too slow for live systems. This is the natural v2+ upgrade for EXHYTE's next-observation planner once the design space grows beyond a discrete menu.](#)

4.3 OPTIMAL BAYESIAN DESIGN FOR MODEL DISCRIMINATION VIA CLASSIFICATION

- **Year:** 2022 (Statistics and Computing).
- **URL:** <https://link.springer.com/article/10.1007/s11222-022-10078-2>SMARTPANTS PRESERVED 3484 | Preprint: <https://arxiv.org/abs/1809.05301>SMARTPANTS PRESERVED 3486SMARTPANTS PRESERVED 3487
- [Hainy et al. choose designs that maximize discriminability between competing models by training a supervised classifier on simulator outputs and using misclassification error as the \(inverse\) design utility — needing far fewer simulations than ABC-based discrimination designs and working with intractable likelihoods. Directly applicable to EXHYTE: pick the diagnostic maneuver whose simulated telemetry best separates fault hypothesis A from B under a classifier — simple to implement with the same ensemble infrastructure.](#)

4.4 BAYESIAN INPUT DESIGN FOR LINEAR DYNAMICAL MODEL DISCRIMINATION

- **Year:** 2020 (Entropy / PMC).
- **URL:** https://pmc.ncbi.nlm.nih.gov/articles/PMC7514835/SMARTPANTS_PRESERVED_3502SMARTPANTS_PRESERVED_3503
- Designs input (excitation) signals that emphasize differences between candidate dynamical models, motivated explicitly by **multiple-model fault detection and isolation**. Precedent that “design the actuation to discriminate fault models” is an established FDI technique — EXHYTE’s diagnostic-maneuver planning is the spacecraft instantiation of this idea.

4.5 BOTORCH — BAYESIAN OPTIMIZATION / ACTIVE LEARNING INFRASTRUCTURE

- **Year:** actively maintained (Meta).
- **URL:** https://botorch.org/docs/multi_objectiveSMARTPANTS_PRESERVED_3520 | Active-learning acquisition source: https://archive.botorch.org/api_modules/botorch/acquisition/bayesian_active_learning.htmlSMARTPANTS_PRESERVED_3522
- Production-grade PyTorch library for Bayesian optimization with acquisition functions including qLogNEHVI/qLogEHVI (noisy, batched, multi-objective) and Bayesian-Active-Learning-by-Disagreement style information-gain acquisitions. In EXHYTE it is less a hypothesis-testing tool than the right engine for *simulation budget allocation* (which parameter settings to simulate next) and for calibrating twin parameters against telemetry via BO.

5. Fault injection frameworks and fault libraries

5.1 NASA VIRTUALADAPT — SIMULATED EPS TESTBED WITH BUILT-IN FAULT INJECTION

- **Year:** open-sourced by NASA; based on the Ames ADAPT testbed.
- **URL:** https://github.com/nasa/VirtualADAPTSMARTPANTS_PRESERVED_3543SMARTPANTS_PRESERVED_3544
- High-fidelity MATLAB/Simulink emulation of the ADAPT electrical power system (batteries, relays, circuit breakers, loads) with a **first-class fault-injection API**: five fault modes — Nominal, Incipient (progressive degradation), Abrupt, Bias, and StuckAt — injectable via GUI, Simulink parameters, or MATLAB scripts. Its fault taxonomy (incipient/abrupt/bias/stuck) is the cleanest published template for EXHYTE’s fault-parameter library, even though the MATLAB dependency argues against direct reuse.

5.2 ADAPT DATASET AND DIAGNOSTIC COMPETITION (DXC)

- **Year:** testbed data 2009–2011, still the canonical EPS diagnosis benchmark.
- **URL:** https://c3.ndc.nasa.gov/dashlink/projects/3/SMARTPANTS_PRESERVED_3561 | https://data.nasa.gov/dataset/adapt-datasetSMARTPANTS_PRESERVED_3563SMARTPANTS_PRESERVED_3564
- Real hardware EPS testbed data where faults were injected physically (disconnecting sources/loads/breakers) and in software via an “Antagonist” layer that blocks commands, sends spurious commands, and corrupts sensor data. Provides labeled real-world fault scenarios (voltages, currents, temperatures, switch states) for validating EXHYTE’s EPS fault reasoning against non-simulated ground truth.

5.3 REACTION-WHEEL FAULT SIMULATION LITERATURE (FRICTION, STICTION, ENCODER FAULTS)

- **Year:** 2022–2024 cluster.
- **URLs:** https://doi.org/10.2514/6.2022-2507SMARTPANTS_PRESERVED_3579 (Basilisk, encoder/RW faults) | https://www.mdpi.com/2075-1702/12/12/856SMARTPANTS_PRESERVED_3581 (LSTM detection with redundant RWs, 2024) | https://papers.phmsociety.org/index.php/ijphm/article/view/4395SMARTPANTS_PRESERVED_3583 (FDI+identification via ML on 3-axis ADCS sim)
- The de facto standard fault parameterization across these papers: multiply nominal **static friction by 5–20x**, add stiction/viscous-friction/resonance modes, randomize injection time, and simulate three-axis-controlled satellites to produce paired nominal/faulty telemetry. This gives EXHYTE a literature-backed prior range for reaction-wheel fault magnitudes and injection protocols — the hypothesis “RW-2 elevated friction” becomes a bounded parameter interval, not a guess.

5.4 OPSSAT-AD — REAL SATELLITE TELEMETRY ANOMALY BENCHMARK (ESA OPS-SAT)

- **Year:** dataset 2024 (Zenodo); benchmark paper 2025 (Scientific Data).
- **URL:** https://www.nature.com/articles/s41597-025-05035-3SMARTPANTS_PRESERVED_3601 | https://zenodo.org/records/12588359SMARTPANTS_PRESERVED_3603 | Code: https://github.com/kplabs-pl/OPS-SAT-ADSMARTPANTS_PRESERVED_3605SMARTPANTS_PRESERVED_3606

- [AI-ready, expert-labeled anomaly dataset from ESA's OPS-SAT CubeSat \(mission ended May 2024\), with baselines from 30 supervised/unsupervised detectors, and realistic operational messiness: data gaps, sampling-rate changes, artifacts, amplitude variation. The natural *real-telemetry* test set for EXHYTE's sim-vs-real comparison layer — and a warning about the preprocessing the twin-comparison pipeline must survive.](#)

5.5 CFS TEST FRAMEWORK (CTF) AND FLIGHT-SOFTWARE FAULT INJECTION

- **Year:** 2021 (NTRS).
- **URL:** https://ntrs.nasa.gov/citations/20210009725SMARTPANTS_PRESERVED_3623SMARTPANTS_PRESERVED_3624
- [NASA's CTF drives automated JSON-scripted test and verification campaigns against cFS flight software targets, and cFS-community practice includes software-implemented fault injection \(message flooding, register/memory corruption\) to validate fault detection and recovery chains. Relevant if/when EXHYTE moves from dynamics-level faults to flight-software-level faults inside a NOS3-class twin.](#)

What this means for our MVP

RECOMMENDED SIMULATOR: BASILISK, WITH NOS3 AS THE LATER HIGH-FIDELITY STAGE

Basilisk is the only candidate that simultaneously satisfies the Testing layer's hard requirements: (1) **Python-native scripting** so the EXHYTE agent can programmatically map a hypothesis to parameter changes (RW friction coefficients, sensor noise/bias, solar panel efficiency) without file-deck manipulation; (2) **faster-than-realtime execution and repeatable Monte Carlo dispersion machinery** for running hypothesis ensembles; (3) coverage of the two subsystems where our first fault library lives — ADCS (reaction wheels with friction/jitter, gyros, star trackers) and power (battery/solar-panel/sink modules); (4) a **permissive ISC license** safe for a commercial product; and (5) published precedent (Carneiro 2022) of Basilisk telemetry with injected reaction-wheel faults training fault-detection models. NASA 42 is comparably rigorous but C-only with socket IPC and a NOSA license; NOS3 is the right *eventual* validation twin (real cFS flight software + COSMOS in the loop) but far too heavyweight for ensemble inference; Orekit only adds value later for precision orbit-determination hypotheses; hapsira/poliastro lack attitude and subsystem models entirely. Thermal is Basilisk's weakest subsystem — for v1, keep thermal faults out of scope or use a simple lumped-parameter thermal model bolted onto the Basilisk ensemble outputs.

RECOMMENDED SBI APPROACH FOR HYPOTHESIS SCORING: AMORTIZED NPE (SBI PACKAGE) FOR FAULT PARAMETERS + NEURAL MODEL COMPARISON FOR HYPOTHESIS SELECTION

Structure the problem in two levels, both trained offline on Basilisk ensembles. *Within* a fault family (e.g., "RW-2 friction elevated"), train amortized **Neural Posterior Estimation** with the `sbi` package (Apache-2.0, actively maintained, embedding networks for raw telemetry windows) to get a posterior over the fault magnitude/onset parameters in milliseconds per new telemetry window — the heat-exchanger condition-monitoring result (82x faster than MCMC with equivalent accuracy) is the direct template. *Across* fault families (competing hypotheses = competing simulator configurations), score hypotheses with **amortized Bayesian model comparison** — a BayesFlow-style neural classifier over the candidate twins yielding posterior model probabilities / Bayes factors, or NRE-based ratios in `sbi`. Use simulation-based calibration and posterior-predictive checks (both built into `sbi`) as the guardrail against overconfident posteriors, and validate the full pipeline on OPSSAT-AD real telemetry and ADAPT EPS fault data before trusting it in the loop.

SIMPLEST VIABLE EXPERIMENT-DESIGN METHOD FOR V1: DISCRETE-MENU EIG VIA NESTED MONTE CARLO (PYRO `CONTRIB.OED`), WITH CLASSIFIER-BASED MODEL DISCRIMINATION AS THE FALLBACK

Do not build policy networks or continuous design optimization for v1. Define a small **discrete design space** — e.g., {continue passive observation, downlink channel set A/B at high rate, command a small diagnostic slew, run a wheel spin-up/spin-down test} — and for each candidate design, estimate **Expected Information Gain about the hypothesis indicator** by reusing the ensembles we already generate: simulate each hypothesis under each design, and compute EIG with Pyro's nested-MC/marginal estimators (or, even simpler and nearly free, the classification approach of Hainy et al. 2022: the design under which a classifier best separates the competing hypotheses' simulated telemetry is the design to run). This needs zero new infrastructure beyond the ensemble runner, is robust with intractable likelihoods, and has a clean upgrade path: Deep Adaptive Design (ICML 2021) amortizes the same objective into a millisecond design policy once v1 proves the loop, and BoTorch takes over if the design space becomes continuous (maneuver magnitude, observation duration).

Part X — Research: Market Landscape

04 — Market Landscape: Commercial Landscape, Customers & YC Positioning (2024–2026)

Research for the **EXHYTE agent**: a closed-loop AI system that ingests spacecraft telemetry, detects anomalies, generates mechanistic root-cause hypotheses via LLMs, and tests them against a digital twin/simulator. Target: satellite operators and mission-ops teams.

Compiled: July 2026. All sources listed in Section 6 were visited via web fetch or returned by web search during this research session.

1. Competitive landscape

1.1 THE MAP

Nobody currently sells the full EXHYTE loop (telemetry → anomaly → LLM mechanistic hypothesis → simulation-verified root cause → recommended action). The market is split into four camps that each own one piece:

<u>Camp</u>	<u>Players</u>	<u>What they own</u>	<u>What they lack</u>
<u>Mission-ops platforms</u> (“Datadog for space”)	<u>Quindar (YC S22), Leanspace, Basalt (YC), Antaris, Kratos OpenSpace, GomSpace HOOP</u>	<u>Telemetry ingest, C2, dashboards, workflow automation</u>	<u>No mechanistic diagnosis; alerts stop at “something deviated”</u>
<u>AI anomaly detection / health</u>	<u>Intella (Milan) — closest competitor, Constellation Space (YC W26), Cognitive Space</u>	<u>Detection, triage, noise reduction, some “root cause analysis” claims</u>	<u>Statistical/ML root-cause, not physics-verified; no closed simulation loop</u>
<u>Digital twin / simulation</u>	<u>Sedaro, Antaris TrueTwin, Ansys/AGI STK</u>	<u>High-fidelity spacecraft models, gov contracts</u>	<u>Twins used for design & training, not wired to live ops diagnosis</u>
<u>Hardware telemetry infra (adjacent)</u>	<u>Sift (\$67M raised, ex-SpaceX), Nominal</u>	<u>AI-ready observability layer for hardware test & ops</u>	<u>Horizontal infra, not a diagnosis product; potential partner or channel</u>

1.2 KEY COMPETITOR DETAIL

- **Quindar (YC S22, Denver)** — “Datadog for Space.” Founded by six ex-OneWeb engineers; SaaS for monitoring/automating satellite fleets so “a single engineer can operate hundreds of satellites.” Raised a **\$6M round in Feb 2024** (KSAT integration) and an **\$18M Series A (Nov 2025)** led by Washington Harbour Partners with Booz Allen Ventures and YC; total ~\$27M. Expanding into **classified government work** with a Denver classified facility — a signal that the money in ops software is flowing toward defense. Quindar is the most likely acquirer-of or competitor-to an EXHYTE wedge; it automates workflow, not diagnosis.
- **Intella (founded 2023, Milan)** — the closest direct competitor. Its **Mercury** platform does AI anomaly detection **plus root-cause analysis** on satellite telemetry, and its CCO publicly argues (SatNews, May 2026) that the real pain is automating *routine nominalconfirmation triage*, not rare-event detection. Partnered with **Telespazio Germany** (EASE-Rise platform) and Digantara. Differentiation for EXHYTE: Mercury’s root-cause is correlational/ML; EXHYTE’s is *mechanistic and simulator-verified* — a testable, falsifiable hypothesis rather than a ranked suspect list.
- **Constellation Space (YC W26, Seattle)** — ex-SpaceX/NASA/Blue Origin founders; “AI operating system for mega-scale satellite networks.” ConstellationOS ingests 100K+ telemetry messages/sec, claims 90%+ failure prediction accuracy and autonomous traffic rerouting in <2s. Raised a **\$6.5M seed (March 2026)**. Still in design-partner phase — proof YC will fund exactly this thesis, and proof the wedge is contested. They focus on *network/link* failures; spacecraft-subsystem root-cause is open.
- **Antaris (\$28M Series A, March 2026, led by WestWave, with Lockheed Martin Ventures)** — “AI-native platform for space missions” built around **TrueTwin** full-mission virtualization and SatOS flight software; claims 2x faster to orbit, 10x cheaper. Antaris proves investors will fund digital-twin-centric space software at Series A scale, but its twin is aimed at design/build/ops virtualization, not closed-loop anomaly diagnosis.

- **Sedaro (Arlington, VA)** — cloud-native digital-twin platform; **SDA Direct-to-Phase-II SBIR (Sept 2024)** to twin GPS and the Proliferated Warfighter Space Architecture, and a **6-year, \$20M AFRL IDIQ** for cloud-native digital twins. Demonstrated on-orbit edge autonomy with NOVI (Feb 2025). Sedaro shows the gov-funded path for twin tech — and is a potential simulation-backend partner rather than a competitor.
- **Sift (LA, ex-SpaceX Dragon/Starlink leads)** — telemetry observability for mission-critical hardware; **\$42M Series B (March 2026)**, StepStone; GV led the \$17.5M Series A); \$67M total. Customers: ULA, Astranis, K2 Space, Parallel Systems, defense programs. Sift is building “the intelligence layer that lets AI interact with hardware” — the strongest evidence that “AI-on-telemetry” is a fundable venture category right now, and the most dangerous well-funded neighbor if it moves from test/observability into on-orbit diagnosis.
- **Cognitive Space (Houston)** — CNTIENT platform, AI-driven constellation tasking/scheduling (mainly EO); ~\$10M total raised; **\$900K NOAA ground-processing demo contract**. Owns planning/scheduling automation, not health diagnosis.
- **Kayhan Space (Boulder)** — Pathfinder spaceflight-safety/conjunction platform; \$12M total raised; 22 operators, 500+ satellites coordinated. Pricing model is instructive: **free “Essentials” tier + paid Pathfinder Pro** — freemium works for wedging into ops teams.
- **Leanspace (Strasbourg)** — API-first “space cloud” for ground/ops software; **€10M Series A (Nov 2025, incl. Capgemini’s ISAI Cap Venture and Qwaltec)**; 20+ spacecraft operators, contracts with Airbus DS, Hispasat, ESA. Evidence that European institutional demand for ops-software modernization is real and growing.
- **Epsilon3 (YC S21, LA)** — “OS for spacecraft and complex operations” (procedures/checklists, not telemetry AI); **\$15M Series A** led by Lux Capital, ~\$75M valuation, 3-dozen+ customers incl. NASA, Blue Origin, Sierra Space. Publishes a public per-seat SaaS pricing page — proof that bottom-up SaaS motion (land on one team, expand) works in space ops.
- **Incumbents:** Kratos **OpenSpace** (software-defined, orchestrated ground systems for defense/SATCOM) and Ansys **STK/AGI** dominate legacy ground/C2 and mission engineering. They move slowly and sell big integrations — classic startup air-cover. **Neuraspace** (Portugal, €25M+ raised) shows the “AI + insurer/regulator angle” applied to collision avoidance rather than health.
- **SpaceKnow** — despite the name, it’s EO imagery analytics (economic intelligence from satellite images), not fleet ops; not a competitor.

Gap statement: every funded player either detects (Intella, Constellation Space), automates workflow (Quindar, Leanspace, Basalt), or simulates (Sedaro, Antaris). No one closes the loop: *hypothesis generation + twin-based falsification + verified root cause*. Academic work is converging on exactly this architecture (see AURA, Section 6 #17), which means the window is open but not for long.

2. YC precedent in space & AI-diagnostics

- **YC space track record:** YC has funded space consistently since W16; top outcomes include **Astranis (W16, ~\$1.2B raised, >\$1B contract backlog, 5 sats on orbit)**, **Stoke Space (\$1.4B)**, **Relativity Space (\$1.4B)**, **Momentus (SPAC)**. 2024–2026 batches: Starcloud (S24, \$170M Series A at \$930M pre — space datacenters), Apolink (F24, \$4.3M seed at \$45M post), Orbital Operations (W25), Cascade Space (X25), Constellation Space (W26, \$6.5M seed), Basalt (“Constellations as a Service,” \$3.5M+ seed led by Initialized). SpaceTech is explicitly named in YC’s recent Requests for Startups.
- **Software-for-space YC comps:** Epsilon3 (S21) and Quindar (S22) both pitched “boring SaaS for space ops” and both raised strong Series A rounds — YC partners have pattern-matched this category successfully twice.
- **AI-for-engineering-diagnostics YC comps (F25/2026 batches):** **Telemetron** (AI-native hardware support — LLMs reasoning over logs, manuals, live telemetry), **Daqstra** (AI orchestration layer for physical R&D test infrastructure in aerospace/energy/robotics), **Deeptrace** (alert triage → root cause → resolution for software on-call). YC is actively funding “LLM reads machine data and diagnoses” across domains; EXHYTE is the spacecraft instance of a thesis YC already believes.

Implication: expect YC to be receptive but to ask “why won’t Constellation Space / Quindar / Intella do this?” — the answer must be the *verified closed loop*, not detection.

3. Market size & dynamics

- **Satellite population is compounding.** ~10,500–14,500 active satellites in early 2026 (tripled since 2019); Starlink alone passed **10,000 active satellites (March 2026)** — roughly two-thirds of everything operational. FCC authorized 7,500 more Gen2 sats in Jan 2026. Projections: 50,000– 100,000+ active satellites by 2030 (Starlink 42K ambition, Kuiper 3,236, China’s Qianfan up to ~14,000).

- **Ops economics force automation.** Traditional GEO: 20–50 dedicated operators per satellite; running Starlink that way would take 200,000–500,000 operators. Illustrative staffing cost curve: ~\$150K/sat/yr at 30 sats manual → ~\$15K/sat/yr at 300 sats with AI → ~\$1.5K/sat/yr at 3,000 sats. Planet’s SkySat team scaled 3→15 satellites with a **3x reduction in operator hours** and eliminated 24/7 staffing by automating anomaly response and routine triage. Manual ops become “unsafe or uneconomical” past ~5–10 satellites (Intella/SatNews). “AI in space operations” projected at ~**22.9% CAGR through 2034**.
- **Ground segment spend:** Novaspace (ex-Euroconsult) forecasts **\$80B cumulative ground-segment market 2023–2032**; GSaaS ~\$400M by 2027 (11% CAGR); satellite ground-station *software* specifically ~**\$1.85B (2024), 8.7% CAGR** — the addressable slice for ops software is ~\$1–2B/yr today and growing with NGSO constellations.
- **Cost of anomalies is existential, and insurers feel it.** ViaSat-3 F1’s antenna anomaly (2023) produced a **\$420M claim — the largest single claim in industry history**; combined with Inmarsat I6-F2’s power anomaly (\$348M cover), Viasat expected ~\$770M in payouts. 2023 was among the worst years ever for space insurance; 2024 losses passed \$500M; several underwriters have exited the market and premiums rose sharply. An agent that shortens anomaly diagnosis, prevents mis-response, and documents root cause has a direct line to insurance economics (premium credits, claims substantiation, loss-adjustment evidence) — Neuraspace already sells adjacent data to insurers.
- **The real daily pain is triage, not detection** (Intella, SatNews 2026): operators drown in “expected, benign, repetitive” threshold alerts, mute alarms, and burn engineer hours confirming nominal behavior across decade-long GEO missions. ESA has even published open benchmarks (OPS-SAT-AD; ESA-ADB) because agency-side anomaly detection remains unsolved at deployment quality — a credibility path and data source for a startup.

4. Buyer analysis

Who owns the pain (ranked for a seed startup):

1. **Small/mid LEO constellation operators (5–100 sats)** — EO, IoT, comms startups past their first launch. Pain onset is sharp at >5 satellites; teams are 3–15 engineers with no dedicated anomaly staff; they buy SaaS fast and have no legacy ground system worth defending. This is the beachhead. (Kayhan reached 22 operators; Leanspace 20+; Quindar landed here first.)
2. **GSaaS / ops-outsourcing providers (KSAT, Telespazio, GomSpace HOOP, Qwaltec)** — they operate *other people’s* fleets and their margin is operator hours; Intella’s Telespazio deal and Quindar’s KSAT integration show they’ll embed third-party AI. One deal = many fleets (channel leverage).
3. **Commercial GEO operators & their insurers** — highest per-anomaly stakes (\$100M+ assets, the ViaSat-3/I6-F2 claims), long procurement but concentrated pain; insurance angle (root-cause evidence, premium reduction) is an unusual wedge.
4. **Defense/civil agencies (SDA, AFRL, NOAA, ESA)** — biggest budgets; enter via SBIR/STTR and SpaceWERX like Sedaro (D2P2 SBIR, \$20M IDIQ) and Cognitive Space (NOAA BAA). Non-dilutive but slow; requires facility clearances at scale (see Quindar building a classified facility).

Procurement realities:

- **ITAR/EAR:** spacecraft ops software can be EAR-classified if genuinely commercial/dual-use, but anything “specially designed” for military spacecraft trips ITAR; penalties up to \$1M per violation. 2024 reforms relaxed licenses for many space items to 40 Wassenaar countries — selling ops software to European/Japanese/Australian operators is now easier than it was. Compliance cost burden falls ~8x harder (as % of foreign sales) on small firms — budget for it from day one, keep the product cloud-hosted and data-classified carefully.
- **Sales cycles:** commercial LEO startups close in weeks–months (Epsilon3’s public per-seat pricing and Kayhan’s freemium tier are proof of PLG-ish motion); primes/agencies take 12–24 months and want on-prem/classified deployments. The pattern that worked (Quindar, Sedaro, Cognitive Space): land commercial SaaS logos → use SBIR/BAA to fund the defense-grade variant → raise Series A on the combined story, often with strategic investors (Booz Allen, Lockheed Martin Ventures, In-Q-Tel-style).
- **What wedge products have worked:** free/cheap tier attached to a daily-use workflow (Kayhan Pathfinder Essentials), procedures/checklists (Epsilon3), monitoring dashboard with automation upsell (Quindar). Wedges that stall: “we’ll replace your whole ground system” (rip-and-replace fear — every incumbent from Kratos down defends this).

5. Adjacent beachheads — is space the right first market?

The same closed-loop architecture (telemetry → LLM hypothesis → twin falsification) applies to:

- **Industrial IoT / AIOps:** AIOps market ~\$10.5–16.6B (2024–25) growing ~18–23% CAGR to \$66B+ by 2035; AI predictive maintenance ~\$12.8B (2025) → ~\$105B (2035). Vastly bigger TAM, but crowded (Datadog, Dynatrace, every PdM vendor) and the digital-twin fidelity that makes verification credible rarely exists for a random factory.
- **Aviation fleet health:** strong twin culture (engine OEMs) but locked up by GE/RR/P&W and certification regimes — brutal for a seed-stage entrant.
- **Robotics/maritime:** the AURA research framework (LLM + digital twin + human-in-loop diagnosis for underwater robotics, arXiv Nov 2025) shows the architecture generalizing — and that academia will commoditize the pattern.

Verdict: space is the right first market. (a) Spacecraft are the one asset class where physics-grade twins already exist or are contractually required (Sedaro/SDA, Antaris TrueTwin); (b) no truck roll is possible — remote diagnosis is the *only* diagnosis, so willingness to pay is structurally higher; (c) anomaly cost per asset is extreme (\$420M single claim); (d) the buyer pool is small enough to enumerate but growing 20%+/yr; (e) YC/VCs have fresh pattern-matches (Sift, Antaris, Quindar, Constellation Space all funded within 8 months). Expand later into defense ground systems, then industrial assets that already have credible twins (energy/grid, launch vehicles).

6. Sources (all visited or returned by search, July 2026)

1. **SpaceNews — “Quindar raises \$18 million to scale satellite operations software” (2025).** <https://spacenews.com/quindar-raises-18-million-to-scale-satellite-operations-software/SMARTPANTS PRESERVED 3996> — Series A led by Washington Harbour Partners with Booz Allen Ventures, FUSE, FCVC, YC; total ~\$27M. Founded 2022 by ex-OneWeb engineers; AI monitoring/automation of satellite ops; proceeds fund a classified Denver facility for government programs. Confirms mission-ops SaaS is Series-A fundable and drifting toward defense revenue.
2. **Y Combinator — Quindar company page, “Datadog for Space” (S22; page current 2026).** <https://www.ycombinator.com/companies/quindarSMARTPANTS PRESERVED 4002> (see also <https://payloadspace.com/a-ga-with-quindar-yc-s22/SMARTPANTS PRESERVED 4004>) — Pitch: single engineer operates hundreds of satellites; monitoring/testing/ops web app. Direct YC precedent for the category and the framing EXHYTE must differentiate from (observability vs. verified diagnosis).
3. **Intella — company site + Mercury product (2025–26).** <https://www.intella.tech/SMARTPANTS PRESERVED 4010> — Milan, founded 2023. Mercury does AI anomaly detection, triage/noise-filtering and root-cause analysis on satellite telemetry; positioned to cut ops cost across constellations. Closest existing competitor to EXHYTE’s detection+diagnosis layers; lacks a simulation-verification loop.
4. **SatNews — Daniele Bellomi (Intella CCO), “Your anomaly detection isn’t the problem” (May 2026).** <https://satnews.com/2026/05/11/your-anomaly-detection-isnt-the-problem/SMARTPANTS PRESERVED 4016> — Argues the cost sink is routine nominal confirmation triage, not rare-event detection; false-positive floods cause alarm muting; manual ops break past 5–10 sats. Cites Planet SkySat: 3x fewer operator hours while assets grew 5x. Defines the pain narrative an EXHYTE pitch should adopt (and shows a competitor already owns part of it).
5. **Y Combinator — Constellation Space company page (W26; fetched 2026).** <https://www.ycombinator.com/companies/constellation-spaceSMARTPANTS PRESERVED 4022> — Ex-SpaceX/NASA/Blue Origin team; ConstellationOS ingests 100K+ telemetry msgs/sec, predicts network failures 5 min ahead (claimed 90%+ accuracy), auto-reroutes in <2s; design-partner stage; \$6.5M seed (March 2026 per Space Ambition). Proves YC is funding this exact thesis *now* — but scoped to link/network failures, leaving subsystem root-cause open.
6. **SpaceNews — “Antaris raises \$28 million Series A to accelerate AI-driven space missions” (March 2026).** <https://spacenews.com/antaris-raises-28-million-series-a-to-accelerate-ai-driven-space-missions/SMARTPANTS PRESERVED 4030> — Led by WestWave Capital with Lockheed Martin Ventures; TrueTwin full-mission virtualization + SatOS flight software; claims up to 2x faster time-to-orbit, 10x cost reduction. Validates digital-twin-first space software at meaningful scale; twin is design/ops-virtualization oriented, not diagnostic.
7. **Sedaro — “Sedaro awarded Phase II SBIR supporting SDA and SSC” (Sept 2024).** <https://www.sedaro.com/news/sedaro-awarded-phase-ii-sbir-supporting-sda-and-sscSMARTPANTS PRESERVED 4036> (see also the \$20M AFRL IDIQ: <https://orangeslices.ai/sedaro-lands-20m-us-air-force-research-laboratory-cloud-native-digital-twins-idig-contract/SMARTPANTS PRESERVED 4038>) — Digital twins of GPS satellites and the SDA PWSA constellation; 6-year cloud-native digital-twin IDIQ with AFRL. Shows the SBIR→IDIQ non-dilutive path for twin tech and a candidate simulation partner/backend for EXHYTE.
8. **SpaceNews — “Sift raises \$42 million in Series B round” (March 2026).** <https://spacenews.com/sift-raises-42-million-in-series-b-round/SMARTPANTS PRESERVED 4044> (see also TechCrunch: <https://techcrunch.com/2026/03/25/with-sift-stack-two-ex-spacex-engineers-are-bringing-the-software-that-helped-launch-rockets-to-the-factory-floor/SMARTPANTS PRESERVED 4046>) — Ex-SpaceX Dragon/Starlink leads; StepStone led; \$67M total; customers ULA, Astranis, K2 Space. “AI infrastructure for mission-critical machines” — the intelligence layer letting AI interact with hardware telemetry. Strongest funding comp for AI-on-telemetry, and a company to partner with or deliberately out-position.

9. **SpaceNews — “Kayhan raises \$7 million and offers autonomous traffic-coordination service” (2023).** <https://spacenews.com/kayhan-raises-7-million-and-offers-autonomous-traffic-coordination-service/SMARTPANTS PRESERVED 4052> — Pathfinder 3.0 autonomous space-traffic coordination; \$12M total; 22 operators / 500+ satellites; free Essentials tier + paid Pro. Demonstrates freemium wedge economics into satellite ops teams.
10. **Cognitive Space — \$4M raise (2023) and NOAA ground-processing contract.** <https://www.cognitivespace.com/news/cognitive-space-raises-4million/SMARTPANTS PRESERVED 4058> and <https://www.cognitivespace.com/news/cognitive-space-wins-NOAA-contract/SMARTPANTS PRESERVED 4060> — CNTIENT AI constellation tasking/scheduling SaaS; ~\$10M total (York IE, Draper); \$900K NOAA BAA demo. Owns planning automation, not health diagnosis; shows civil-agency BAA as an early revenue path.
11. **SpaceNews — “Epsilon3 raises \$15 million for space project management platform” (2023).** <https://spacenews.com/epsilon3-raises-15-million-for-space-project-management-platform/SMARTPANTS PRESERVED 4066> (pricing: <https://www.epsilon3.io/pricingSMARTPANTS PRESERVED 4068>) — YC S21; Lux Capital-led Series A; ~\$75M valuation; 3-dozen customers incl. NASA, Blue Origin, Sierra Space; public SaaS pricing. Proof of bottom-up SaaS motion in space ops.
12. **SpaceNews — “Leanspace raises €10 million Series A” (Nov 2025).** <https://spacenews.com/leanspace-raises-e10-million-series-a-round-to-bring-software-defined-satellite-operations-to-enterprise-and-institutional-space-programs/SMARTPANTS PRESERVED 4074> — API-first satellite-ops cloud; investors incl. Capgemini’s ISAI Cap Venture and Qwaltec; 20+ operators; contracts with Airbus DS, Hispasat, ESA. Confirms European institutional demand for software-defined ops.
13. **Space Ambition (Substack) — “Y Combinator Bets on New Space Startups, 2024-2026 Edition” (2026).** <https://spaceambition.substack.com/p/y-combinator-bets-on-new-space-startups-623SMARTPANTS PRESERVED 4080> — Catalogs YC space bets: Starcloud (S24, \$170M A at \$930M pre), Apolink (F24, \$4.3M), Constellation Space (W26, \$6.5M), Basalt, etc.; top YC space outcomes Stoke (\$1.4B), Relativity (\$1.4B), Astranis (\$1.2B). SpaceTech named among top YC RFS categories — timing is favorable.
14. **Y Combinator — Astranis company page (W16) + funding history (2026).** <https://www.ycombinator.com/companies/astranisSMARTPANTS PRESERVED 4088> (see also <https://research.contrary.com/company/astranisSMARTPANTS PRESERVED 4090>) — From YC prototype to >\$1B valuation, \$1.2B raised (\$300M Series E, May 2026, a16z/BlackRock), 5 satellites on orbit, >\$1B backlog. The canonical “YC hard-tech space” outcome investors will anchor on.
15. **New Space Economy — “AI as Mission Control: How Autonomous Satellite Operations Are Changing the Ground Segment” (March 2026).** <https://newspaceeconomy.ca/2026/03/30/ai-as-mission-control-how-autonomous-satellite-operations-are-changing-the-ground-segment/SMARTPANTS PRESERVED 4096> — Traditional GEO needed 20-50 operators/satellite; Starlink at traditional ratios would need 200,000-500,000 operators. Cost curve: \$150K→\$15K→\$1.5K per sat/yr as AI extends the ratio. Names the emerging standalone “intelligence layer” in ground segment; AI-in-space-ops ~22.9% CAGR through 2034.
16. **Orbital Radar / SpaceNexus / KeepTrack — satellite population counts (2026).** <https://orbitalradar.com/how-many-satellites-in-orbitSMARTPANTS PRESERVED 4102>, <https://spacenexus.us/blog/how-many-satellites-in-space-2026SMARTPANTS PRESERVED 4104>, <https://keeptrack.space/x-report/spacex-brief-2026-03-18SMARTPANTS PRESERVED 4106> — ~10,500-14,500 active satellites early 2026 (3x since 2019); Starlink passed 10,000 active (March 2026), ~two-thirds of all operational sats; FCC authorized +7,500 Gen2 (Jan 2026); 50K-100K+ projected by 2030 (Kuiper 3,236; Qianfan up to ~14,000). The structural demand driver.
17. **Via Satellite — “Viasat to Initiate Service on ViaSat-3 Satellite, Starts Receiving Insurance Money” (Feb 2024).** <https://www.satellitetoday.com/finance/2024/02/07/viasat-to-initiate-service-on-viasat-3-satellite-starts-receiving-insurance-money/SMARTPANTS PRESERVED 4112> (context: <https://www.datacenterdynamics.com/en/analysis/paying-the-premium-why-2023-was-a-bad-year-for-space-insurance/SMARTPANTS PRESERVED 4114>) — ViaSat-3 F1 antenna anomaly: \$420M claim, the largest single claim in industry history; with Inmarsat I6-F2’s power anomaly, ~\$770M expected payouts. 2023 among the worst space-insurance years on record; 2024 losses >\$500M; underwriters exiting, premiums rising. Quantifies the cost-of-anomaly and the insurance-aligned buyer.
18. **Novaspace (Euroconsult) — “Ground segment market forecast to reach cumulative value of \$80 billion by 2032” (2023).** <https://nova.space/press-release/soaring-ground-segment-market-forecast-to-reach-cumulative-value-of-80-billion-by-2032/SMARTPANTS PRESERVED 4120> (software slice: <https://dataintel.com/report/satellite-ground-station-software-marketSMARTPANTS PRESERVED 4122>) — \$80B cumulative ground-segment spend 2023-2032, NGSO-driven; GSaaS → ~\$400M by 2027 (11% CAGR); ground-station software ~\$1.85B in 2024 at 8.7% CAGR. Bounds the near-term SAM for ops software.

19. **arXiv — “AURA: A Collaborative Reasoning Framework for Anomaly Diagnostics in Underwater Robotics” (Nov 2025).** https://arxiv.org/html/2511.03075v1SMARTPANTS_PRESERVED_4130 (see also constrained-LLM+DT anomaly detection: https://arxiv.org/pdf/2604.03790SMARTPANTS_PRESERVED_4132) — LLM + high-fidelity digital twin + human-in-the-loop: a low-level agent translates telemetry into a structured problem description; a high-level diagnostic agent reasons to root cause. This is the EXHYTE architecture, published for underwater robotics — evidence of technical feasibility and that the pattern will commoditize; speed to a spacecraft-specific product matters.
20. **Nature Scientific Data — “The OPS-SAT benchmark for detecting anomalies in satellite telemetry” (2025).** https://www.nature.com/articles/s41597-025-05035-3SMARTPANTS_PRESERVED_4140 (also ESA-ADB benchmark study: https://arxiv.org/html/2607.07335SMARTPANTS_PRESERVED_4142) — ESA released real CubeSat telemetry anomaly benchmarks with baselines from 30 ML algorithms; follow-up mission (OPS-SAT VOLT) flies to validate detectors. Public, credible training/eval data for a demo, plus proof agencies consider deployed-grade anomaly detection unsolved.
21. **SpaceNexus — “ITAR and EAR Compliance: What Every Space Startup Needs to Know” (2025/26).** https://spacenexus.us/blog/itar-ear-compliance-space-startupsSMARTPANTS_PRESERVED_4148 (2024 reform detail: https://www.hklaw.com/en/insights/publications/2024/11/new-export-control-rules-present-key-regulatory-changes-for-spaceSMARTPANTS_PRESERVED_4150) — “Specially designed” for military use → ITAR (fines to \$1M/violation); commercial dual-use → EAR. 2024 reforms removed license requirements for many space items to 40 Wassenaar countries, easing allied-market sales. Compliance burden is ~8x heavier on small firms as % of foreign sales — plan architecture and hiring around it early.
22. **GMI / MAK Data Insights — AIops & AI predictive-maintenance market sizing (2025).** https://www.gminsights.com/industry-analysis/aiops-marketSMARTPANTS_PRESERVED_4156 and https://www.makdatainsights.com/reports/global-ai-based-predictive-maintenance-marketSMARTPANTS_PRESERVED_4158 — AIops ~\$10.5-16.6B (2024-25) → \$66B+ by 2035 (~18-23% CAGR); AI-based predictive maintenance \$12.8B (2025) → \$105.6B (2035). Frames the later expansion story: same closed-loop diagnostics engine, orders-of-magnitude larger adjacent TAM.
23. **Forbes — “The Top Startups To Watch From Y Combinator’s Fall 2025 Batch” (Nov 2025).** https://www.forbes.com/sites/dariashunina/2025/11/13/the-top-startups-to-watch-from-y-combinators-fall-2025-batchSMARTPANTS_PRESERVED_4164 — Highlights Telemetron (LLM-based hardware diagnostics/support over logs, manuals, live telemetry) and peers like Dagstra (AI orchestration of aerospace/energy test infrastructure) and Deepttrace (alert→root-cause→resolution for software on-call). YC is systematically funding LLM-driven diagnostics across engineering domains — EXHYTE fits an active YC pattern.
24. **Kratos — “OpenSpace Platform Supporting Dynamic, Software-Defined Satellite Ground Systems” (2020, current product line).** https://www.kratosdefense.com/newsroom/kratos-introduces-openspace-platform-supporting-dynamic-software-defined-satellite-ground-systemsSMARTPANTS_PRESERVED_4170 — The incumbent benchmark: virtualized, orchestrated ground systems for defense/SATCOM (SpectralNet, quantum products, OpenSpace Platform). Defines what “enterprise-grade” means to GEO/defense buyers and why startups win on intelligence layer, not infrastructure replacement.

(Additional context sources surfaced during search: Quindar \$6M/KSAT — *Via Satellite* Feb 2024, https://www.satellitetoday.com/finance/2024/02/06/quindar-secures-6m-in-funding-for-spacecraft-management-ai-toolsSMARTPANTS_PRESERVED_4176 ; Basalt seed — *TechCrunch* May 2024, https://techcrunch.com/2024/05/30/basalt-closes-3-5m-to-turn-space-into-a-software-defined-industrySMARTPANTS_PRESERVED_4178 ; Neuraspace €25M and insurer/regulator customers — https://www.neuraspace.com/SMARTPANTS_PRESERVED_4180 and https://blog.neuraspace.com/request-for-quotation-for-a-space-tracking-telescope-0SMARTPANTS_PRESERVED_4182 ; Telespazio-Intella-Digantara EASE-Rise partnership — https://www.satelliteevolution.com/post/telespazio-partners-with-digantara-and-intella-to-deliver-next-generation-satellite-operations-throuSMARTPANTS_PRESERVED_4184 ; Planet SkySat ops automation — *SmallSat* 2019, https://digitalcommons.usu.edu/smallsat/2019/all2019/86/SMARTPANTS_PRESERVED_4186 .)

7. YC positioning recommendation

Sharpest one-sentence pitch: “Satellite fleets triple every five years but anomaly response is still a war room — EXHYTE is the AI flight-controller that detects an anomaly, hypothesizes the physical root cause, and *proves it against a digital twin* before an operator touches a console.”

(Alternative framing if partners push back on scope: “Verified root-cause-as-a-service for spacecraft — we don’t just flag anomalies, we close them.”)

Likely first customer profile: A LEO constellation operator with **5-50 satellites and a 3-15 person ops team** (EO/IoT/comms startup, 2-5 years post first launch) that just muted half its alerts; secondarily a **GSaaS / ops-outsourcing provider** (KSAT-, Telespazio-, GomSpace-HOOP-type) whose margin is operator hours and who will embed EXHYTE across

many customer fleets. Defense (SDA/AFRL SBIR) is the year-2 non-dilutive layer, not the first sale.

Main objections investors will raise (prepare answers):

1. "Isn't this Quindar/Intella/Constellation Space?" — They stop at detection/workflow; nobody verifies mechanistic hypotheses against a simulator; the closed loop is the moat and the wedge.
2. "Where does the digital twin come from?" — Building per-customer high-fidelity twins is service-shaped, not SaaS-shaped. Need an answer: auto-calibrated reduced-order twins from telemetry
 - partnerships (Sedaro/Basilisk/42/open-source sims), and a fidelity-tiered product.
3. "TAM: only ~70 meaningful commercial operators; ground software is ~\$2B/yr." — Counter with constellation growth (10K→50K+ sats), per-anomaly stakes (\$420M single claim), insurance-aligned pricing, and the industrial-diagnostics expansion (AIOps/PdM \$100B+ by 2035).
4. "LLM hallucination in safety-critical ops / will operators trust it?" — The twin is the hallucination filter: every hypothesis must reproduce the telemetry signature in simulation before surfacing; human-on-the-loop by default (mirrors current agency practice).
5. "Data access and ITAR." — Start with allied commercial operators under EAR-friendly 2024 rules; use ESA's public OPS-SAT/ESA-ADB benchmarks to derisk before customer data lands.

Strongest demo ideas for a YC interview (pick 2):

1. **Live replay of a real ESA anomaly:** stream OPS-SAT benchmark telemetry, have EXHYTE detect the anomaly, generate 3 ranked mechanistic hypotheses (e.g., EPS degradation vs. sensor fault vs. thermal), run each in the twin, and show one hypothesis reproducing the exact telemetry signature — detection-to-verified-root-cause in under 5 minutes, next to the days/weeks a human anomaly-review board takes.
2. **The "muted alarm" demo:** feed a day of noisy multi-satellite telemetry with hundreds of threshold alerts; show EXHYTE triaging 98% as benign-with-explanation and escalating the two real faults with twin-verified causes — directly attacking the triage pain Intella's own CCO says is the real cost sink.
3. **Fleet-scale counterfactual:** take a public failure narrative (ViaSat-3-class antenna/power anomaly), reconstruct it in simulation, and show EXHYTE's hypothesis loop isolating the fault class from downlink telemetry alone — then show the insurance framing: "this is what a \$420M claim looks like when you can prove root cause in hours."

Part XI — Session Handoff

HANDOFF — 2026-07-19

Done

- **Source paper** read + text-extracted ([EXHYTE-research-doc.pdf](#) , [docs/research/exhyte-paper-extracted.txt](#)).
- **borromeanRings governs the repo**: [borromeanrings.toml](#) tuned (python/ [spacethink](#) /src layout; hygiene requires [README/SPEC/pyproject](#)). Gate green (all 7 checks) — run the [verify.sh](#) script from your local borromeanRings install.
- **Literature review, 90+ verified sources**: [docs/research/01...04-*.md](#) (anomaly detection SOTA, LLM discovery agents, digital twins + SBI, market/YC landscape) and the synthesis [docs/research/literature-review.md](#) .
- **SPEC.md v0.2** (CS130 style): user stories + ACs, quality-attribute scenarios, module decomposition, all 7 scope decisions finalized (OPS-SAT-AD→ESA-ADB; Basilisk; ADCS/reaction wheels; Telemanom-lineage detector; [sbi](#) NPE + model comparison; discrete-menu EIG; role-separated LLM layer with causal gate).
- **docs/IMPLEMENTATION PLAN.md**: Phases 0-5 with exit criteria, two-founder track split, dependency graph.
- **docs/MVP ROADMAP.md**: pitch, 2 demo scripts, 12-week timeline, success metrics, GTM, investor objections + answers.
- **Code**: TDD scaffold — [src/spacethink/domain](#) immutable entities ([Hypothesis/EventOfInterest/FaultParameter/SimMapping](#)) with construction-time validation; 8 tests green.

Pending (next session, in order)

1. Phase 0 remainder: [runstore/](#) (fs+SQLite artifact store) + [spacethink](#) CLI entry point.
2. Phase 1 start: OPS-SAT-AD download + [ingest/](#) _parser (dataset links in [docs/research/01-...md](#) §7).
3. Phase 2 start (parallel track): Basilisk install + minimal 4-wheel scenario + RW friction fault injection (precedent: Carneiro 2022, linked in [docs/research/03-...md](#) §3.6).
4. [git init](#) + first commit — repo is not yet under version control (deliberately left for founder decision on hosting/license).
5. Founder review of SPEC §8 decisions and MVP ROADMAP pitch — these encode judgment calls the co-founders should ratify.

Failed / blocked

- Nothing failed. No blockers. (PDF page rendering unavailable — [poppler-utils](#) not installed — text extraction via [pypdf](#) used instead; no information lost that matters.)

Key context for whoever picks this up

- Top technical risk is sim-to-real alignment (SPEC §9 R1); the plan de-risks via synthetic-first eval + OPS-SAT-AD/ADAPT real-data validation + calibration checks in CI.
- Never report point-adjusted F1 anywhere — evaluation rigor is a stated differentiator.
- The gate is the definition of done. Run it before claiming anything works.